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**VCaiLuong: Microservices System for the Development and  
Preservation of Cai Luong's Intangible Cultural and Artistic  
Heritage**

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**VCaiLuong: Virtual Museum for the Development and  
Preservation of Cai Luong's Intangible Cultural and Artistic  
Heritage**

APPROVED BY:

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THESIS COMMITTEE

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# Abstract

In the contemporary era of economic development and global integration, where monetary value and economic benefit are often prioritized, cultural and spiritual values are increasingly marginalized, particularly among younger generations. Among the most affected is *Cai Luong*, a traditional Vietnamese theatrical art form, which is experiencing a decline in popularity due to cultural diversification and limited accessibility for youth. In response, this study proposes a technological solution to support the preservation and promotion of traditional arts in the digital age through the development of the **VCaiLuong system**.

VCaiLuong is a digital system designed as a virtual cultural ecosystem to store, present cultural values associated with *Cai Luong* and other traditional Vietnamese art forms, contributing to cultural preservation and development. Key system features are categorized into two main applications: the Visitor Application features, including virtual museums, watch party functions for cultural content, and digital artifact storage, offer new and accessible pathways for younger audiences to engage with these heritage elements; and the Admin Portal Application features, encompassing content contribution, monitoring tools, and creator tools. These features establish a comprehensive flow to preserve and contribute to cultural development from the perspectives of both the visitor and the content creator.

This research first outlines the necessity of employing digital technology in the conservation of intangible cultural heritage, using *Cai Luong* as a focal case supported by an analytical table. Comparative analyses of existing systems, such as the Digital Heritage Collection (DHC) by TUVU Communications, the virtual museum platform ArtSteps, and the Vietnam National Fine Arts Museum (VNFAM) virtual museum, are presented to assess their capabilities, strengths, and limitations, which inform the design of VCaiLuong.

The VCaiLuong system is constructed using contemporary system design principles, including cloud computing, microservice architecture, caching, and modular application separation. It comprises two core applications: a Visitor application and an Admin Portal application, each serving distinct functional roles. Technologies such as ReactJS, TailwindCSS, NestJS (TypeScript), Java, Spring Boot, MySQL, Redis, Azure Blob Storage, and cloud services from Microsoft Azure and Firebase are employed to ensure system scalability, maintainability, and high performance, especially in managing 3D assets and digital records.

The implementation of the system offers interactive virtual museum exploration, real-time Watch Party viewing of cultural content, user engagement tools such as chat, notifications, and bookings, alongside robust content creator tools including a museum creation editor platform, a user management platform, and functionality to customize museum based on design. Furthermore, performance is enhanced, and resource usage is optimized through the introduction of automation via background jobs and scheduling mechanisms within the microservice environment. The parallel processing paradigm is also applied, utilizing the job queue pattern to efficiently handle long and resource-intensive tasks.

This study concludes with a critical evaluation of VCaiLuong's design objectives, strengths, and limitations. A comparison with DHC, ArtSteps, and VNFAM highlights VCaiLuong's innovations and areas for future improvement. Lastly, the research outlines directions for future development to further strengthen VCaiLuong's role as a sustainable platform for the preservation and promotion of *Cai Luong* in the digital era.

# Glossary

|                |                                               |
|----------------|-----------------------------------------------|
| <b>UI</b>      | User Interface                                |
| <b>DHC</b>     | Digital Heritage Collection                   |
| <b>VNFAM</b>   | Vietnam National Fine Arts Museum             |
| <b>API</b>     | Application Programming Interface             |
| <b>PUB/SUB</b> | Publish or Subscribe                          |
| <b>SMS</b>     | Short Message Service                         |
| <b>2D</b>      | Two Dimension                                 |
| <b>3D</b>      | Three Dimension                               |
| <b>VICAST</b>  | Vietnam Institute of Culture, Arts and Sports |
| <b>DDD</b>     | Domain-Driven Design                          |

# Chapter 1

## INTRODUCTION

### 1.1 The necessity of preserving Cai Luong cultural heritage by information technology

#### 1.1.1 What is CaiLuong cultural heritage and preservation by technology?

Cai Luong, a form of modern folk opera originating in southern Vietnam, is a unique and significant aspect of Vietnamese cultural heritage, which not only played an important role during the resistance period but is also a spiritual nourishment for the Vietnamese people. It combines traditional music, spoken drama, and contemporary themes, reflecting the social and emotional life of the Vietnamese people. However, like many traditional art forms, Cai Luong faces the risk of fading due to changing entertainment preferences, urbanization, and generational gaps. In response to these challenges, information technology has become a crucial tool for the preservation and revitalization of Cai Luong culture.

However, in the face of globalization and rapid societal change, many intangible cultural heritage forms, including Vietnam's Cai Luong, a modern folk opera blending traditional music, spoken drama, and contemporary themes, are at risk of disappearing. Once central to Vietnamese cultural identity, Cai Luong has seen a sharp decline due to urbanization, shifting entertainment preferences, and generational divides. Simultaneously, the tourism industry is transitioning from mass tourism to experiential tourism, as modern travelers seek authentic, participatory cultural experiences [1]. This trend has given rise to "prosumers," tourists who actively co-create their experiences [2], highlighting the need for innovative solutions to revitalize heritage forms like Cai Luong in ways that resonate with contemporary audiences.

**The mission of preservation through information technology** aims to create an endeavor to safeguard the cultural heritage of Cai Luong and related art forms by prioritizing the archiving, documentation, and exhibition of Cai Luong artifacts, including costumes, scripts, and related conventional art forms. According to VICAS, in the current integrated society, by applying advanced technology, preservation facilitates the transformation of live performances, traditional treasures, and related cultural items into curated digital records to create a platform that can promote the essence of Cai Luong to new generations and our community. Additionally, these digital archives serve as a repository, ensuring researchers, artists, and the community can have an opportunity to investigate and provide educational outreach with a large source of information and the vibrant culture of Cai Luong. Beyond mere protection, the preservation via technology seeks to raise the connection between Cai Luong and traditional art forms with new generations by embedding cultural elements into modern applications, which are populated in modern life, particularly among younger generations who may be unfamiliar with traditional art forms. Finally, the preservation through information technology aims to enhance cultural literacy, inspire renewed appreciation for Cai Luong, and ensure its influence with the Vietnamese community and among international visitors.

### 1.1.2 Some Approaches to Safeguarding Cai Luong Cultural Heritage Using Technology.

Cai Luong Preservation through technology can be implemented in various ways, all aimed at maintaining original values and reflecting those values in contemporary life. However, there are two methods that have been utilized for various traditional art forms that can both safeguard cultural traditions and encourage their appreciation among younger generations: Digital Archiving, and Digitization of live performance into digital platforms.

In this context, digital technologies, particularly Artificial Intelligence (AI), Virtual Reality (VR), and Augmented Reality (AR), emerged as powerful tools to recreate, disseminate, and enhance heritage experiences [6]. The intersection of the challenges in preserving Cai Luong, the evolving expectations of tourists, and the potential of digital technology raises an urgent question: how can technology be applied to both preserve the core cultural values of Cai Luong and create attractive cultural experiences for contemporary audiences?

Digital Archiving is an effective approach to preserving Cai Luong cultural heritage based on the development of virtual museums. This method requires digitizing historical costumes, scripts, musical instruments, and other related items associated with Cai Luong, transforming them into high-resolution digital records, such as images, videos, and three-dimensional models. These digital records can then be used to present within the virtual museum, which not only ensures the long-term preservation of fragile artifacts against physical deterioration but also helps to enhance accessibility for a large community.

Digitizing of Cai Luong Performances on Digital Platforms involves recording live Cai Luong performances in high-quality audio-visual formats, capturing the nuances of acting, singing, and musical accompaniment. This method not only preserves the ephemeral nature of live performances for future generations but also facilitates interactive features like subtitles, commentary, and virtual reality experiences to enhance user interaction.

## 1.2 Problem Statement

In recent years, the popularity of Cai Luong, a traditional art form of Vietnamese culture, has been steadily declining, particularly among younger generations. While it has historically been cherished and widely performed, Cai Luong today mainly appeals to older generations, causing a demographic imbalance that raises concerns about its long-term sustainability and relevance in contemporary Vietnamese society.

According to a 2014 survey conducted by the Can Tho College of Culture and Arts, only 6.1% of Vietnamese youth participants expressed an interest in traditional musical genres, including Cai Luong, Tuong, and Cheo. Even fewer, just 4.3% showed enthusiasm for traditional theatrical performances, including Cai Luong dramas[1]. These statistics demonstrate the growing disconnect between audiences and traditional performing arts. The limited participation with younger audiences threatens the preservation and continuity of the Cai Luong art form in modern society. Furthermore, the decline in the younger audience also creates a looming human resource crisis in the field, as fewer young people are inspired or trained to become artists, directors, writers, or technicians involved in Cai Luong. Without a new generation to inherit and contribute to the tradition, both the creative and operational ecosystems that sustain Cai Luong may deteriorate, accelerating its marginalization in Vietnam's cultural landscape.

Alongside decreasing audience numbers, both live attendance and digital viewership for Cai Luong performances have significantly decreased. This has resulted in fewer new



productions being staged and a reduction in the diversity of performances. Moreover, a critical issue is the lack of comprehensive preservation mechanisms for Cai Luong’s cultural artifacts. Scripts, costumes, recordings, and other historical materials are scattered, poorly maintained, or absent from any centralized archive, leading to the gradual loss of invaluable cultural heritage. This problem is compounded by the absence of a systematic approach to digital archiving, which could ensure the long-term protection and accessibility of these resources [2].

Furthermore, many Cai Luong theaters and institutions have yet to embrace digital transformation in meaningful ways. Where digital initiatives exist, they are often limited in scope or temporary, failing to engage younger audiences who increasingly consume culture through modern technologies. The absence of effective online platforms and digital tools restricts the ability of Cai Luong to reach a broader and more diverse audience. Additionally, these technological gaps hinder the art form’s ability to evolve and adapt to changing social contexts, which is crucial for its survival in a fast-paced, digitized world.

While various digital applications have emerged to support learning and entertainment across multiple fields, Cai Luong’s integration into these platforms remains minimal. Most existing efforts focus on passive content delivery rather than interactive engagement, which limits their appeal and effectiveness. The need for a comprehensive solution that combines digital archiving, user-friendly access, and community engagement is evident. Such an approach would not only safeguard Cai Luong’s rich heritage but also revitalize interest among younger generations by making the art form more accessible and relevant.

In conclusion, Cai Luong faces significant challenges due to demographic shifts, lack of preservation infrastructure, and insufficient digital integration. Without immediate and sustained efforts to address these issues, there is a serious risk that this traditional art form will continue to decline, ultimately leading to the loss of an important cultural legacy. Therefore, developing sustainable digital preservation and promotion strategies is essential to ensuring that Cai Luong remains a living, evolving part of Vietnamese culture for future generations.

### 1.3 Scope And Objectives

The observable decline in public interest, particularly among younger demographics, poses a significant existential threat to traditional performing arts, specifically Cai Luong. In direct response to this cultural imperilment, the VCaiLuong system is formally proposed as a comprehensive digital intervention utilizing modern information technologies to execute four core functions: archiving, exhibiting, revitalization, and creation. The project is fundamentally designed as an integrated application combining the structural components of a virtual museum with the interactive capabilities of a digital performance space, thereby securing both the conservation of heritage and contemporary audience engagement. Crucially, moving beyond the preservation mandate, the VCaiLuong system is strategically intended to establish an interactive creative platform. This platform transitions the system from a passive custodial role to a dynamic environment that fosters experimentation among artists and the public, utilizing digital tools to manipulate and re-interpret the artistic elements of Cai Luong.

The scope of the proposed system encompasses two primary components. The first component is the Virtual Cai Luong Application, which functions as a centralized repository for digitized cultural artifacts, including costumes, scripts, musical instruments, photographs, and historical documents. All materials within this repository are organized following established metadata standards to ensure accessibility, efficient retrieval,

and long-term digital preservation. In addition to archival functions, the application provides interactive “meeting-room” performance environments, where traditional Cai Luong performances are presented in live or recorded formats. These virtual spaces enable audiences to attend remotely, engage in real-time discussions, and interact with other viewers, thereby replicating the communal atmosphere characteristic of live theater experiences.

The second component is a Digital Content Management System, designed to support the aggregation, organization, and dissemination of traditional Cai Luong performances and related digital assets. This system facilitates the integration of cultural materials into a creative digital platform, allowing users, particularly content creators—to construct personalized virtual exhibition spaces. Within these spaces, users can present their interpretations of traditional culture, manage their curated collections, and design interactive cultural experiences for public engagement.

### 1.3.1 Goals of VCaiLuong

The primary objective of VCaiLuong is to preserve and promote Cai Luong through an accessible, user-friendly, and content-rich digital platform. The system seeks to ensure that cultural heritage is not lost to time but is instead revitalized through digital documentation, community interaction, and immersive experiences. This is achieved through the digitization of artifacts, the transformation of live performances into online experiences, and the use of technology to reach broader, younger, and more diverse audiences.

The platform also aims to become a long-term digital archive and educational tool, helping students, researchers, and artists explore the values and techniques of Cai Luong. By promoting data-driven cultural strategies, the platform will empower institutions to adapt to audience needs and foster sustainable cultural growth.

### 1.3.2 System Actors and Basic Functions

The VCaiLuong system incorporates three primary categories of system actors: **end-users, content providers, and system administrators**. Each actor performs a critical function within the system, ensuring the integrity, accessibility, and sustainability of the digital platform.

**End-users include a broad spectrum of individuals such as students, researchers, cultural enthusiasts, and tourists.** These users interact with the system primarily through two core modules: the virtual museum and the interactive performance space. The virtual museum provides access to curated exhibits featuring digitized Cai Luong artifacts, including costumes, scripts, photographs, and performance archives. Meanwhile, the performance module simulates a live theater experience, where users can attend or rewatch Cai Luong performances in a virtual meeting room format. This feature also facilitates user interaction and commentary, thereby enhancing audience engagement and fostering cultural dialogue.

**Content providers consist of recognized Cai Luong institutions, theater groups, and independent artists.** These stakeholders are granted access to content management tools that allow them to upload, organize, and curate their own digital exhibitions and performances. By enabling theaters and individuals to create and maintain their own cultural showcases, the platform decentralizes access to the digital preservation process and encourages a more dynamic and participatory model of cultural transmission. Furthermore, this functionality allows institutions to expand their reach, engage with digital audiences, and maintain relevance in an increasingly digital society.

**System administrators** are responsible for overseeing the overall operation and governance of the platform. Their duties include content moderation, user management, security enforcement, and granting authors the appropriate access rights to create and modify their virtual exhibits. Administrators also monitor the performance of the virtual spaces, resolve technical issues, and coordinate with content providers to ensure alignment with the platform's preservation objectives.

This structured distribution of roles facilitates a clear separation of responsibilities, allowing for efficient operation and sustainable growth. By integrating technical, cultural, and administrative actors into a cohesive system, the application aims to preserve, present, and promote Cai Luong in a manner that is both inclusive and adaptable to modern technological standards.

# Chapter 2

## BACKGROUND AND RELATED WORKS

### 2.1 Background

#### 2.1.1 The Importance of Preserving Traditional Art Forms

Traditional art forms are core values of cultural identity, history, and spiritual values of a nation. They not only represent intangible heritage inherited through many generations, encompassing language, customs, music, performance, and storytelling. In many countries, including Vietnam, these art forms have played an essential role in promoting community cohesion and promoting moral values.

However, in the face of rapid globalization, urbanization, and the digital transformation of society, many traditional art forms are at risk of being forgotten and replaced. Some factors, like declining public interest, lack of formal education in traditional arts, aging artist populations, and insufficient preservation infrastructure, lead to this erosion. Younger generations are increasingly detached from these cultural roots, often viewing traditional performances as outdated or irrelevant in the modern world.

Therefore, preserving traditional art forms is not only an urgent action at present but also a strategy to maintain identity and cultural diversity of each country. In addition, the safeguarding of traditional arts aligns with UNESCO's convention for Safeguarding of the Intangible Cultural Heritage (2003) [3], which emphasizes the importance of community involvement and intergenerational transmission in cultural preservation. This necessitates the development of modern, technology-enabled strategies that can revitalize public interest, enhance accessibility, and ensure long-term sustainability.

#### 2.1.2 Global Efforts in Preserving Traditional Art Forms

Across the world, various countries have recognized the importance and urgency of preserving traditional art forms in the context of modernization and rapid digital transformation. Japan stands as a pioneer in the preservation of classical performing arts, particularly Kabuki and Noh. These art forms are not only safeguarded through national broadcasting programs and digital archiving but also incorporated into state-supported national theater systems. In addition, Japan has adopted advanced technologies such as online performances, three-dimensional simulations, and digital repositories in national museums, facilitating public access and intergenerational transmission of traditional knowledge [?].

Additionally, South Korea is not standing outside the effort to protect traditional artistic values by implementing several innovative initiatives to preserve its art forms, especially Pansori. In recent years, cultural institutions and research centers have applied artificial intelligence (AI) to restore and enhance the quality of historical recordings. The South Korean government has invested in the development of web-based and mobile platforms to digitize and disseminate traditional content to a younger audience, thereby fostering a stronger connection between cultural heritage and modern digital consumers [?].

In Vietnam, traditional art forms such as Quan ho, Ca tru, and Cheo have received recognition from UNESCO as Intangible Cultural Heritage, which has raised public awareness of their cultural value. Community-driven initiatives, including the production of educational videos and the organization of local festivals, have contributed to their preservation. However, the application of information technology in these efforts remains limited, lacking the comprehensive and systematic digital infrastructure needed for large-scale and sustainable preservation.

From these global examples, it is evident that integrating information technology with traditional preservation methods has become an essential trend and an urgent action. Integration not only enhanced accessibility and engagement, especially among younger generations, but also ensured the long-term sustainability of intangible cultural heritage in the context of societal and technological change

### **2.1.3 The role of information technology in cultural preservation**

Information technology (IT) has emerged as a powerful enabler in the preservation and dissemination of traditional cultural arts. Its integration into cultural heritage efforts has provided new, innovative pathways for documenting, sharing, and revitalizing intangible art forms that are otherwise at risk of being forgotten.

In addition to preservation, IT facilitates user-friendly access to traditional art through the development of web and mobile applications. These platforms allow broader audiences, especially younger generations, to explore and interact with traditional content in more modern and engaging ways. Educational apps, streaming services, and interactive websites make it easier to promote cultural appreciation beyond physical and temporal constraints.

Moreover, emerging technologies such as virtual reality (VR), three-dimensional simulations, and artificial intelligence (AI) have transformed the way cultural experiences are presented. These technologies can recreate traditional performance environments, simulate stage settings, and even reproduce lost or damaged performances. Such immersive experiences enable users to feel a deeper connection with cultural heritage, regardless of their geographical location.

In summary, information technology not only strengthens preservation efforts but also enhances the visibility, accessibility, and sustainability of traditional cultural arts in the digital era.

### **2.1.4 Cai Luong in the Context of Cultural Preservation**

Cai Luong is a distinctive form of Vietnamese traditional theater that blends elements of singing, acting, and instrumental music. Originating in the early 20th century, particularly in Southern Vietnam, Cai Luong evolved as a modernized version of classical Vietnamese opera and quickly gained popularity among the masses for its emotional storytelling and cultural resonance. For decades, it served as an important cultural medium reflecting social values, historical narratives, and everyday life, firmly embedding itself in the identity of Vietnamese performing arts.

Despite its historical and artistic value, Cai Luong has experienced a significant decline in popularity, especially among younger generations. Once a central form of entertainment in Southern Vietnam, it now struggles to retain public interest in an age dominated by digital media and globalized pop culture. The audience for Cai Luong has aged, and few

new artists are emerging to carry the tradition forward.

Current efforts to preserve Cai Luong remain fragmented and limited. Some television programs and YouTube channels continue to broadcast performances, but these efforts are sporadic and lack a coordinated or systematic digital preservation strategy. There is a notable absence of interactive platforms where younger audiences can engage with Cai Luong in ways that resonate with their digital lifestyles. Without such engagement, the art form risks further marginalization and eventual obsolescence.

Applying information technology to the preservation of Cai Luong is both timely and essential. Modern communication tools such as digital storytelling, online performance platforms, and social media offer the potential to revitalize the art form and make it more accessible and attractive to contemporary audiences. Through digitization, immersive technologies, and interactive educational platforms, Cai Luong can be reintroduced to younger generations.

## 2.2 Related works

This literature review explores recent research on the preservation of traditional art forms through digital platforms and virtual art systems. It focuses on the implications of emerging technologies for safeguarding and revitalizing intangible cultural heritage, particularly in the context of digitization and increased public engagement in the digital age.

### 2.2.1 Digital Heritage Collection of TUVAC Communications

**a) Overview:** The Digital Heritage Collection is designed to effectively support the preservation, documentation, and dissemination of Vietnam’s diverse cultural heritage. This digital platform aims to provide core functionalities such as the collection and archiving of cultural artifacts and intangible heritage in various media formats, which contribute to the community-driven content through web applications and mobile interfaces. By promoting and fostering collaboration among local communities, cultural experts, and the public, the system aims to enhance cultural awareness and engagement while ensuring the integrity and accessibility of heritage data. DHC meets the critical needs of cultural preservation management by providing a transparent, sustainable, and user-friendly environment that promotes the safeguarding and revitalization of traditional arts and practices.

#### **b) Some Basic Functions of the System:**

**Knowledge acquisition:** A fundamental function of the DHC platform is to facilitate the acquisition and dissemination of knowledge regarding Vietnam’s intangible cultural heritage (Fig. 2.1). The platform provides users with a wide range of educational resources in many multimedia formats, including photographs, hand-drawn illustrations, short documentary films, and audio files. These resources comprehensively document the diverse cultural practices, traditional arts, and oral histories of various ethnic groups and regions within Vietnam. By offering open access to these educational materials, DHC serves as a valuable reference for researchers, educators, students, and the broader public interested in cultural heritage. This function thus plays a pivotal role in raising awareness and fostering understanding of Vietnam’s cultural diversity.

**Content contribution:** In addition to its role as an educational repository, the DHC platform functions as a participatory digital archive, allowing community members and cultural experts to contribute content actively (Fig. 2.2). This open-access model encour-

ages individuals and groups to publish and share various forms of cultural documentation, such as images, videos, narratives, and descriptions of artistic practices. By decentralizing content contribution, the platform ensures inclusivity and diversity in cultural representation, empowering local communities to participate directly in heritage preservation.

**Cooperation and Development:** The DHC platform also fosters cooperation and cultural development by facilitating the organization and dissemination of various heritage-related activities (Fig. 2.3). These include press conferences, online discussions, storytelling sessions, educational webinars, and virtual competitions aimed at promoting cultural heritage by providing a digital space.

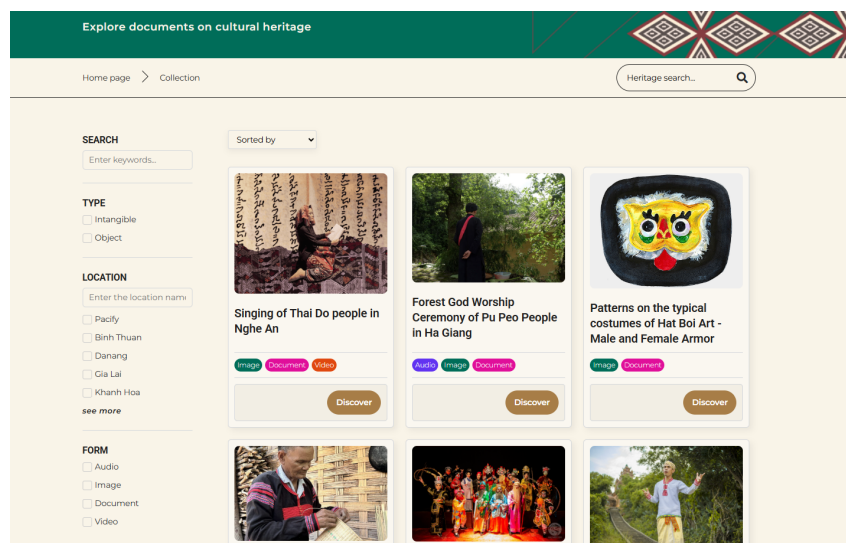


Figure 2.1: Documentation interface of the DHC platform.

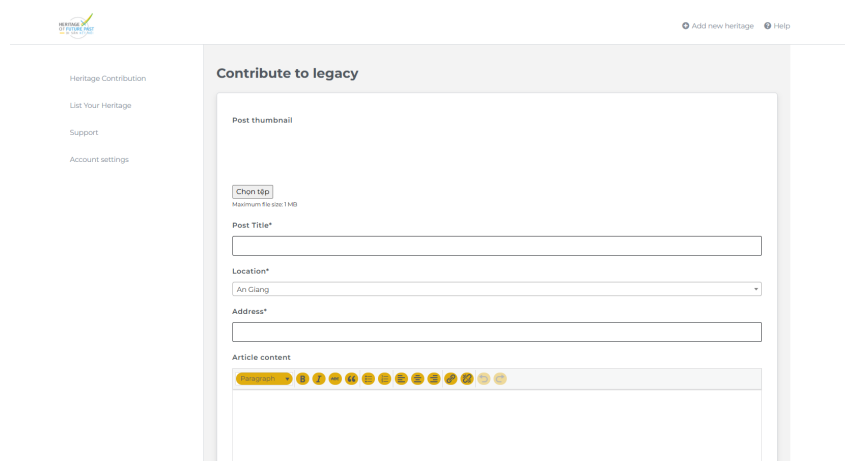


Figure 2.2: Community contribution interface of the DHC platform.

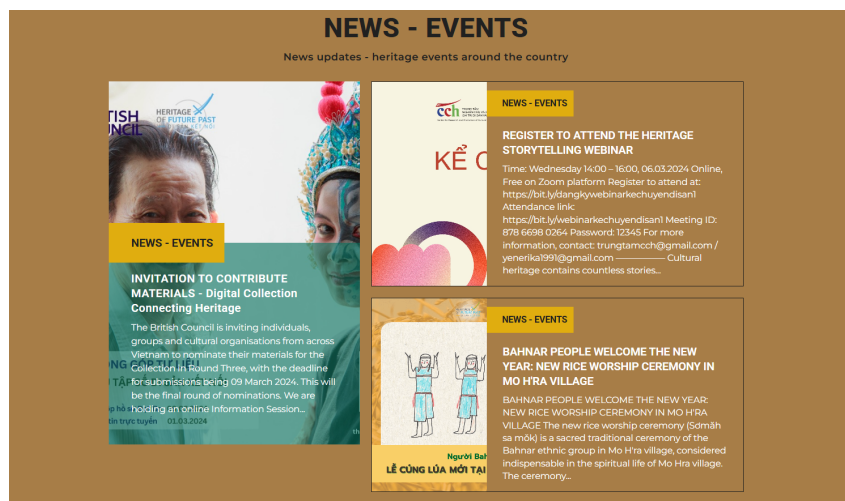


Figure 2.3: Event-sharing and cultural storytelling space on DHC

### c) Advantages:

The DHC platform offers a range of technological advantages that contribute to its effectiveness in preserving and disseminating Vietnam's intangible cultural heritage. One of its primary strengths is its open-access architecture, which enables a broad range of users, including community members, researchers, and cultural practitioners, to access, read, and publish documentation freely. This openness fosters active participation and democratizes cultural heritage preservation, moving beyond traditional top-down models where only experts control content.

The platform supports a diverse array of digital content formats, including images, videos, audio recordings, and text documents. This multi-format capability is crucial for capturing the multifaceted nature of intangible heritage, including oral histories, traditional music, rituals, and visual arts. By accommodating various media types, the platform allows for richer, more immersive documentation that reflects the complexity of cultural expressions.

Additionally, an important technological contribution is the platform's facilitation of community collaboration and knowledge exchange. Through its support of virtual conferences, workshops, and interactive sessions, the system encourages users to share insights, discuss challenges, and collaboratively develop solutions related to heritage preservation. These functionalities transform the platform from a static repository into a living, dynamic network where cultural knowledge can evolve and be revitalized.

From a system architecture perspective, the platform incorporates basic but effective content management features, including metadata tagging, categorization, and search capabilities. These features enhance the discoverability and organization of heritage materials, which is essential for users seeking specific information or conducting academic research. Additionally, the platform's web-based nature ensures relatively wide accessibility, compatible with various devices and operating conditions across regions, including rural areas.

### d) Limitations and areas for improvement:

Despite its valuable contributions, the DHC also exhibits some technical and design challenges that cause inconvenience in use and need to be addressed to improve its overall impact and usability.

The UI and experience currently reflect a design approach more suitable for academic



or archival use rather than for broader public engagement. The interface tends to be text-heavy and document-centric, which may discourage or complicate participation by younger users or those with limited digital literacy skills. The complex or outdated UI can reduce user retention and limit the platform’s effectiveness in reaching and engaging diverse community groups. Modern UI/UX principles, including more visual and interactive design elements, could make the platform more approachable and user-friendly.

The open contribution model, introduces risks related to content quality and authenticity. Without a rigorous moderation or peer-review mechanism, the platform may become susceptible to the submission of inaccurate, incomplete, or unverifiable information. This is particularly critical in the field of cultural heritage, where the accuracy of recorded traditions, stories, and practices is essential to maintaining cultural integrity.

From a technical infrastructure standpoint, the platform’s reliance on server-side rendering and traditional document storage may constrain its scalability and responsiveness, especially as multimedia content grows in volume and size. Large audio and video files can slow down content delivery, negatively affecting user experience. Furthermore, the platform’s current focus on minority cultural preservation through an academic lens may inadvertently limit accessibility and engagement from non-academic users within these communities. Simplifying contribution workflows, supporting local languages or dialects, and incorporating storytelling tools tailored for grassroots users could help bridge this gap.

## 2.2.2 Vietnam National Fine Arts Museum Virtual Tour

### a) Overview:

The three-dimensional virtual tour platform of the Vietnam Fine Arts Museum (VN-FAM) represents a significant advancement in the application of digital technologies for cultural heritage preservation in Vietnam. Designed to address the increasing demand for remote accessibility and digital transformation within the museum sector, the platform enables users to engage with cultural artifacts and exhibition spaces through an immersive, web-based virtual environment.

In addition, the system employs advanced 3D rendering technologies to create an interactive virtual space where users can explore and engage directly with artifacts. It supports the integration of multimedia content, such as textual annotations, audio narratives, and video materials, directly into the virtual galleries. This approach aligns with international practices in virtual museum development and provides an enriched user experience that supports both cultural education and public engagement. As such, the VNFAM virtual tour platform contributes meaningfully to national efforts in digital heritage preservation and the dissemination of Vietnamese fine arts to a broader, global audience.

### b) Some basic functions:

The Vietnam Fine Arts Museum’s 3D virtual tour platform incorporates a range of functional components that directly support the long-term preservation, promotion, and accessibility of national cultural heritage through the application of digital technology.

**Virtual Spatial Navigation in 3D:** The system allows users to explore a digitally reconstructed version of the museum space through 360-degree panoramic navigation (Fig. 2.4). This interactive spatial experience simulates physical movement within the museum, ensuring that the cultural significance and curatorial layout of the physical space are preserved and communicated in digital form. This function is particularly valuable in maintaining access to heritage collections during periods of physical inaccessibility, such

as during pandemics or for international audiences.

**Digitally Augmented Exhibits with Multimedia Integration:** The virtual museum provides various artifacts related to fine arts in Vietnam. Each artifact within the virtual tour is linked to contextualized multimedia content, including high-resolution images, curatorial texts, audio guides, and video documentation (Fig. 2.5). This feature fosters a deeper understanding and more meaningful engagement with the artworks, helping to preserve intangible heritage such as stories, interpretations, and oral traditions.

**Multilingual Content Delivery:** To facilitate broader access and promote intercultural understanding, the platform supports both Vietnamese and English (Fig. 2.5). This bilingual interface ensures that cultural knowledge embedded within Vietnam’s artistic heritage can be transmitted to both domestic and international communities, reinforcing the museum’s role in global heritage preservation.

**Thematic Curation and Navigable Collections:** Artworks are grouped by historical periods, artistic styles, or cultural themes, allowing users to browse according to specific interests. This function not only improves navigation but also supports structured cultural education and the thematic preservation of Vietnamese fine arts traditions.

These features and their implementations correspond directly to the outcomes presented in Table 2.1.

| System Functionality               | Technological Implementation                                        | Intended Outcome                                                                  |
|------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 3D Virtual Navigation              | Panoramic 360-degree rendering using WebGL and virtual camera logic | Simulates real-world museum experience; increases accessibility for remote users  |
| Multimedia-Enhanced Exhibits       | Embedded text, image, audio, and video content linked to artifacts  | Enhances educational value and user engagement; supports diverse learning styles  |
| Bilingual Content Support          | Dual-language UI and content encoding                               | Expands reach to international audiences; supports cultural diplomacy             |
| Cross-Device Compatibility         | Responsive web design and browser-agnostic architecture             | Ensures seamless access from desktops, tablets, and mobile devices                |
| Thematic Categorization and Search | Metadata tagging and categorized routing                            | Facilitates efficient content discovery and personalized user experience          |
| Server-Side Rendering Architecture | Backend-rendered scenes and preloading optimizations                | Ensures stability and performance, especially for large-scale media assets        |
| Interactive Object Exploration     | Clickable objects with linked information layers                    | Encourages exploration and inquiry; supports contextual understanding of artworks |

Table 2.1: Mapping of System Functionalities to Cultural Preservation Outcomes



Figure 2.4: Virtual Museum Platform of VNFAM.



Figure 2.5: Artifact Description of VNFAM.

### c) Advantages:

The virtual museum platform offers numerous benefits to both the institution and its wide-ranging audience. One of the most significant advantages is increased accessibility. By providing a virtual format, the system enables users worldwide to explore the museum's collections without the constraints of physical location or travel, thereby democratizing access to cultural heritage.

Moreover, the integration of multimedia content, including images, videos, and audio descriptions, substantially enhances heritage education. This multi-modal approach supports diverse learning preferences and fosters a deeper appreciation of art and culture among younger generations and students. It transforms the museum experience into a more engaging and informative interaction, bridging the gap between traditional cultural education and modern digital consumption.

In addition to education, the platform plays a critical role in cultural preservation. By digitizing valuable artworks and collections, the system helps protect fragile physical artifacts from deterioration and compensates for limitations in physical exhibition space. This digital preservation ensures the longevity and wider dissemination of cultural heritage assets.

Finally, the platform's interactive features encourage public engagement. By enabling users to explore artifacts and exhibitions in an immersive environment, the system makes art and history more approachable, stimulating curiosity and fostering a stronger connection between the public and cultural heritage.

#### **d) Limitations and areas for improvement:**

Despite these strengths, the platform faces several technical and usability challenges that impact its overall effectiveness. Currently, the user experience is relatively static; the absence of advanced interactive features such as guided tours, real-time narration, or community feedback mechanisms limits the depth of user engagement. Such dynamic functionalities are increasingly common in virtual heritage platforms and play a critical role in sustaining user interest.

The interface design also presents limitations. It remains predominantly document-focused with a traditional layout, which may not resonate with younger users or those accustomed to contemporary, visually rich digital interfaces. This outdated design can hinder intuitive navigation and detract from the overall user experience.

From a technical standpoint, the platform’s server-side rendering model may cause performance bottlenecks when handling large multimedia files, such as high-resolution images or videos, especially during periods of high user traffic. This can lead to slower loading times and a less seamless experience for users.

### **2.2.3 ArtSteps - VR Exhibitions and 3D Exhibit Creation Tools**

#### **a) Overview:**

The VR exhibition platform, equipped with integrated creation tools, provides a comprehensive functional model that supports the complete workflow from cultural content producers to end users. It addresses the need for a unified digital environment where authorized creators can construct virtual exhibition spaces tailored to specific cultural or artistic themes. This platform is particularly valuable in contexts requiring accessible and interactive digital representations of cultural heritage.

From a technical standpoint, the platform’s server-side rendering model may cause performance bottlenecks when handling large multimedia files, such as high-resolution images or videos, especially during periods of high user traffic. This can lead to slower loading times and a less seamless experience for users. Moreover, the reliance on real-time rendering at the server level increases computational overhead, which may strain system resources when multiple users simultaneously request complex scenes or interact with 3D environments. Without adequate caching strategies or content delivery network (CDN) integration, the system may also face latency issues, particularly for users accessing the platform from geographically distant regions. **b) Some basic functions:**

**Virtual Museum Exploration:** Exhibitions are presented as fully navigable 3D virtual museum spaces, where artifacts are organized into categories such as object type, cultural theme, or historical context. Users can move freely within the virtual environment, interact with individual items, and access detailed information about each asset. This function enhances engagement, supports immersive exploration, and enables users to experience the spatial and contextual relationships between artifacts, simulating the feeling of a physical museum visit (Fig. 2.6).

**Content Creation and Sharing Tools:** ArtSteps integrates a built-in 3D studio that allows creators to design personalized virtual museums. Users can import diverse media types—including images, videos, 3D models, and descriptive text—and arrange them within interactive exhibition spaces. Once completed, these virtual museums can be shared with the community, facilitating collaborative cultural preservation, promoting creative participation, and expanding public access to digitally preserved heritage (Fig. 2.7).

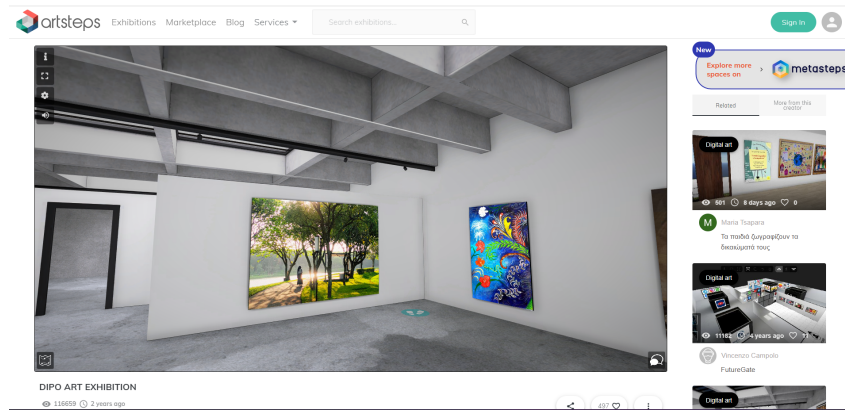


Figure 2.6: Virtual Museum of ArtSteps.

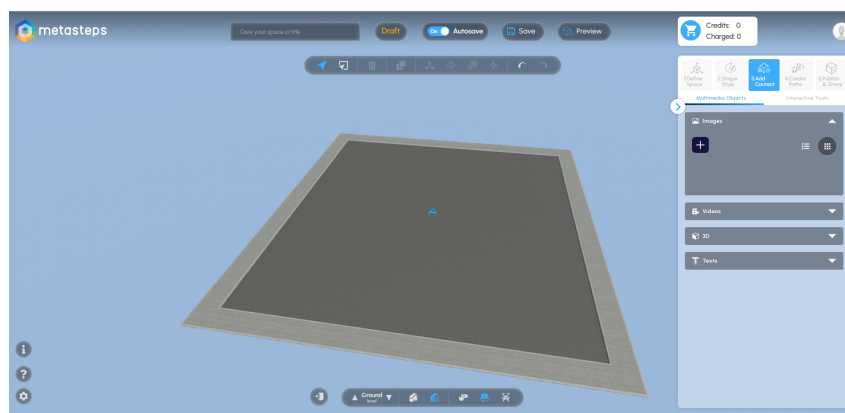


Figure 2.7: Virtual Museum Creation Platform of ArtSteps.

### c) Advantages:

ArtSteps offers a diverse content-sharing platform that allows users to explore a wide variety of exhibitions from multiple virtual museums. The platform's interactive navigation and game-like controls make it particularly engaging for younger audiences, enhancing both exploration and learning. Additionally, ArtSteps employs a modern, user-friendly interface that is designed not only for cultural presentation but also for educational purposes. The system enforces standardized asset formats and rules, ensuring that created content maintains high quality and consistency across exhibitions.

Regarding the content creation platform, ArtSteps integrates a fully-featured 3D editor within the application, providing creators with versatile tools to design their own virtual museum templates. This functionality enables the system to host multiple museums with distinct layouts and designs, supporting a rich diversity of virtual spaces. Creators can incorporate a wide range of asset types, including images, videos, and especially 3D models, which significantly enhances the variety and interactivity of each exhibition. As a result, the platform facilitates engaging, immersive, and educational experiences for users while promoting high-quality content creation within the community.

### d) Limitations and Areas for Improvement:

Despite its advantages, ArtSteps has several limitations that affect both content creators and end users. For creation, access can be challenging for users without prior design experience, particularly because the platform does not provide extensive pre-defined tem-

plates to guide exhibition layout. Additionally, the tools available within the creator platform are relatively limited compared to professional 3D modeling software, such as Blender, which restricts the complexity and flexibility of virtual exhibits.

Furthermore, the integration of the 3D editor directly within the application introduces performance challenges, especially when handling large or complex 3D assets. These limitations may impact system responsiveness and user experience, particularly on lower-end devices. Addressing these issues, through enhanced templates, more advanced creation tools, and performance optimizations, represents key areas for future improvement to make the platform more accessible, versatile, and efficient for a broader range of users.

## 2.2.4 Significance of the Research for Building VCaiLuong

The purpose of this research is to conduct a comparative analysis of three related applications - DHC, VNFAM, and ArtSteps, to evaluate their potential contributions to the development of the VCaiLuong system. This section mainly focuses on examining and comparing the features, functionalities, and advantages of these systems, especially in their contribution to the preservation and promotion of cultural heritage.

### Research Questions:

1. What are the key features and functionalities of the DHC, VNFAM, and ArtSteps systems in supporting the preservation and promotion of cultural heritage?
2. How do the technological implementations of DHC, VNFAM, and ArtSteps contribute to accessibility, education, and community involvement in heritage preservation?
3. What are the strengths and limitations of the two systems from an information technology perspective in contributing to the development of the VCaiLuong system?

### Principal Findings:

The findings of this research indicate that all the **Digital Heritage Collection (DHC)**, the **Vietnam Fine Arts Museum Virtual Tour (VNFAM)** and the **ArtSteps - VR Exhibitions and 3D Exhibit Creation Tools** systems employ a diverse range of technological methods and functional features aimed at enhancing accessibility and safeguarding cultural heritage. Despite their shared objective, the three platforms adopt distinct methodologies and technical approaches in how they implement digital preservation and user engagement. Table 2.2 below shows these key differences.

The comparative study of the Digital Heritage Collection (DHC), the Vietnam Fine Arts Museum Virtual Tour (VNFAM), and ArtSteps - VR Exhibitions and 3D Exhibit Creation Tools systems yields several key findings that directly inform the technical and functional design of the proposed VCaiLuong system. These findings highlight how different approaches to cultural preservation through information technology can be strategically applied to support VCaiLuong, the digital safeguarding and promotion of Cai Luong, a traditional Vietnamese opera.

| <b>DHC System</b>                                                            | <b>VNFAM System</b>                                                            | <b>ArtSteps System</b>                                                              |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Designed for the preservation of intangible heritage (rituals, oral history) | Focused on digitizing tangible cultural assets (paintings, sculptures, spaces) | Focused on 3D virtual exhibitions and interactive content creation                  |
| Open-access platform allowing user contributions and community publishing    | Curated experience maintained by institutional staff                           | Community-driven platform enabling creators to design and share virtual museums     |
| Supports multiple file types (text, audio, image, video)                     | Integrates multimedia (audio, video, text) into interactive 3D virtual space   | Supports images, videos, 3D models, and text with interactive navigation            |
| Encourages public discourse through virtual events and collaborative tools   | Offers immersive museum-like experience through 360° navigation                | Provides game-like navigation, asset interaction, and immersive VR-like exploration |
| Utilizes traditional web-based UI, suitable for archival browsing            | Employs panoramic rendering and virtual camera logic for spatial interaction   | Modern UI designed for both cultural presentation and educational engagement        |
| Lacks strict content moderation tools, which may affect authenticity         | Controlled curation ensures accuracy and artistic integrity                    | Enforces asset format rules to maintain quality while allowing creative freedom     |
| Lower performance with multimedia-heavy files due to server-side rendering   | Optimized for rendering visual scenes but lacks advanced interactivity         | Integrated 3D editor may face performance challenges with large or complex assets   |

Table 2.2: Comparison of Features and Functionalities in DHC, VNFAM, and ArtSteps Systems

### Summary:

This comparative study highlights the complementary strengths of the DHC, VNFAM, and ArtSteps platforms in the digital preservation of cultural heritage. DHC demonstrates excellence in community-driven documentation and supports a wide range of media formats, creating a dynamic archive of living heritage. VNFAM, on the other hand, offers a highly immersive, museum-like experience that facilitates cultural education and appreciation through interactive technologies. ArtSteps contributes additional value by enabling creators to design and share fully interactive 3D virtual exhibitions, combining engagement with creative expression.

Despite their respective strengths, each platform has notable limitations. DHC suffers from an outdated interface and lacks robust content validation mechanisms, while VNFAM provides a more static user experience with limited opportunities for community participation. ArtSteps, although highly interactive and creative, can be difficult to access for users without design experience, limiting its usability for a broader audience. These observations suggest that an effective system, such as VCaiLuong, should integrate the advantages of all three platforms, combining contribution-driven functionality, immersive visualization, and accessible creation tools to support both cultural preservation and active community engagement.

# Chapter 3

## METHODOLOGY

This chapter will focus on the system’s structure, its key components, and the interactions between them. The system’s design will be discussed in several sections, each covering an essential aspect of VCaiLuong architecture and functionality.

### 3.1 System Overview

#### 3.1.1 Project overview and scope

The VCaiLuong system functions as a virtual museum, providing an interactive space where visitors can experience live performances, learn about the history of Cai Luong, and engage with multimedia content related to traditional Vietnamese art forms. This platform offers an immersive experience, allowing users to interact with digital exhibitions, view performances, and explore multimedia content that highlights the beauty and historical significance of these cultural traditions. The primary objective of VCaiLuong is to leverage technology to make traditional art forms accessible to a global audience, ensuring their preservation for future generations while maintaining their relevance in the modern world.

Visitors can immerse themselves in a diverse range of multimedia exhibits, including video performances, historical articles, and audio recordings. These exhibits provide an in-depth exploration of Cai Luong, its evolution, and its pivotal role in Vietnamese culture. Video performances allow visitors to watch both live and digitized traditional performances, while historical articles and educational content offer valuable insights into the cultural significance of the art form. In addition to these static exhibits, the virtual museum incorporates interactive elements, such as digitalized artifacts, which visitors can engage with for a more enriched learning experience. For example, users can explore historical scripts, traditional musical instruments, or costumes associated with Cai Luong, providing a hands-on approach to understanding the cultural heritage of Vietnam.

Moreover, VCaiLuong is not a static platform. It offers the capability to host watch parties and community events, enabling visitors to experience live performances together and enhance communication with fellow users. These interactive features, including in-room chatting, museum chat boxes, and watch party functionalities, are designed to foster a sense of community and encourage active engagement. By enabling such communication, the platform enriches the user experience in ways that traditional museum visits cannot, allowing performers and audiences to connect and collaborate across geographical boundaries. This dynamic feature ensures that the platform not only serves as an educational tool but also creates a space for cultural exchange and interaction among a global audience.

Beyond the visitor-focused features, the VCaiLuong system also provides a comprehensive suite of tools for administrators and content creators, enabling them to curate, customize, and manage their own digital museums. Through an integrated creator dashboard, authorized users can upload and organize artifacts in multiple formats, including images, videos, audio clips, 3D objects, and textual documents, ensuring that diverse



forms of cultural heritage can be accurately represented within the platform. In addition to managing individual artifacts, creators can design and personalize museum templates, construct thematic exhibition halls, and structure narrative pathways that guide visitors through specific cultural stories or historical periods. The administrative interface further allows creators to manage community, facing activities such as watch parties, where they can schedule and broadcast cultural heritage videos, particularly Cai Luong performances, to broad audiences. These capabilities not only empower cultural practitioners, researchers, and institutions to disseminate their work but also foster a dynamic ecosystem where new digital exhibitions and interactive experiences can be continuously developed and shared with the public.

### 3.1.2 Actor and Their Roles

In the VCaiLuong platform, there are three primary actors with distinct roles: Visitor Users, Author Users, and Admin Users. Each actor interacts with the system in different ways, contributing not only to the platform’s overall functionality, but also to the preservation of Cai Luong and other traditional art forms.

**Visitor Users:** Visitor users are the primary audience of the VCaiLuong platform. They are individuals who join the platform to explore and learn about Cai Luong and other traditional Vietnamese art forms. These users access the virtual museum to view multimedia exhibits, watch video performances, read historical articles, and engage with educational content about Cai Luong. The platform allows them to interact with various elements of the museum, such as digital artifacts, musical instruments, and costumes, enhancing their learning experience.

Visitor users can participate in interactive features like watch parties, where they can view live or digitized performances alongside other users, and engage in real-time communication through chat functions. This interaction fosters a community aspect, allowing visitors to share experiences and knowledge. The primary role of the visitor is to consume content, participate in museum rooms or community events, and learn about the cultural significance of Cai Luong while enjoying the immersive features offered by the platform.

**Author Users:** Author users are responsible for creating, customizing, and managing cultural content within the VCaiLuong platform. They are granted permissions to upload multimedia artifacts in various formats, such as images, videos, audio recordings, textual documents, and 3D models, to construct meaningful digital exhibits. Author users can design and personalize their own museum spaces by selecting templates, organizing exhibition layouts, and curating thematic collections that highlight specific aspects of Cai Luong or other Vietnamese cultural traditions. Moreover, they can initiate community-oriented activities such as watch parties, where they share cultural heritage videos and performances with the public. Through these capabilities, author users contribute directly to the continuous enrichment of the platform’s digital cultural archive.

**Admin Users:** Admin users hold the highest level of authority within the VCaiLuong system, overseeing platform management, stability, and adherence to cultural preservation objectives. They review and approve content submitted by authors, grant and manage user permissions, and supervise the overall structure of digital museums. Admins ensure exhibits and performances remain accurate, relevant, and engaging while maintaining content diversity by introducing new categories and updating default templates for creators. They monitor system performance, track user engagement, and implement technical optimizations to enhance usability and reliability. Additionally, admins coordinate large-

scale community activities such as cultural discussions, live-streamed performances, and platform-wide watch parties, sustaining VCaiLuong’s operational integrity and continuously enhancing its features to support a vibrant, accessible, and educational environment dedicated to preserving and promoting Cai Luong and other Vietnamese traditional arts.

### 3.1.3 Technique and Tools

The VCaiLuong platform leverages a robust set of modern technologies and tools to deliver an interactive, scalable, and efficient virtual museum experience. The following outlines the key tools and techniques used across version control, front-end development, back-end development, database management, and cloud services.

#### a) Version Control:

Git is employed as a decentralized version control system to track changes in the source code throughout the development of VCaiLuong. It enables seamless collaboration among multiple developers, ensuring efficient management of code versions and facilitating integration of new features and updates [4]. Additionally, GitHub Actions is used as a CI/CD pipeline tool to automatically detect changes in the release branch. When new changes are pushed, it sets up the required environment, runs tests, and deploys the latest version of the platform, ensuring smooth and consistent updates while minimizing manual intervention.

#### b) Front-end Development:

The front-end of VCaiLuong is built to provide an engaging and interactive user experience, utilizing the following technologies:

**TypeScript:** A superset of JavaScript, TypeScript adds static typing to enhance code reliability and maintainability, enabling developers to catch errors early and improve the development process for complex front-end components [5].

**React.js :** An open-source JavaScript library developed by Facebook, React.js simplifies the creation of dynamic and interactive user interfaces, forming the core of VCaiLuong’s front-end architecture [6].

**Redux:** A state management library, Redux uses actions to ensure consistent and predictable updates to the application’s state, serving as a centralized store for managing the platform’s complex state interactions [7].

**Zustand:** A lightweight and scalable state management library, Zustand provides a simpler alternative to Redux, allowing developers to manage component state efficiently with minimal boilerplate, improving responsiveness and maintainability [?].

**Axios:** A promise-based HTTP client for JavaScript, Axios facilitates communication between the front-end and back-end services, enabling secure and efficient API calls to fetch, update, and manage data in VCaiLuong [?].

**Tailwind CSS:** A utility-based CSS framework, Tailwind CSS provides predefined classes to efficiently style HTML elements, enabling rapid and consistent design implementation across VCaiLuong’s interface [8].

**@react-three/fiber:** A React renderer for Three.js, this library enables the integration of 3D graphics and animations into the VCaiLuong platform, enhancing the immersive experience of virtual exhibits. [9]

**@react-three/drei:** A collection of reusable components and helpers for @react-three/fiber, drei simplifies the implementation of advanced 3D features, such as interactive artifacts and virtual environments [10].

**@react-three/rapier:** A physics engine integration for @react-three/fiber, rapier

adds realistic physics simulations to interactive 3D elements, enriching user interactions with digitalized artifacts like traditional instruments and costumes [11].

**Shadcn/ui:** Shadcn/ui is a library of React components designed to be both accessible and customizable, leveraging Tailwind CSS for styling [12].

### c) Back-end Development:

The back-end of VCaiLuong is designed to ensure high performance, scalability, and flexibility, utilizing the following technologies:

**TypeScript:** Used in the back-end to enhance code quality and maintainability, TypeScript ensures robust type-checking and error detection during development [5].

**NestJS:** A progressive Node.js framework, NestJS provides a modular and scalable architecture for building server-side applications, leveraging TypeScript for structured and maintainable code [13].

**Node.js :** An open-source, cross-platform runtime environment, Node.js enables efficient execution of JavaScript outside the browser, powered by Google Chrome’s V8 engine for high performance [14].

**Java and Spring Boot:** Java, combined with the Spring Boot framework, is used to develop robust and scalable back-end services, providing enterprise-grade features for handling complex business logic and integration [15].

**BullMQ:** A Node.js library for handling distributed job queues, BullMQ enables the back-end to process background tasks, schedule notifications, and manage asynchronous workflows efficiently, ensuring reliable execution of time-sensitive operations [?]

### d) Database:

The VCaiLuong platform employs efficient database systems to manage and store large volumes of multimedia content and user data:

**MySQL:** A relational database management system, MySQL is used for structured data storage, ensuring reliable and efficient management of user information, exhibiting metadata, and other relational data [16].

**Redis:** An in-memory data structure store, Redis serves as a high-performance caching layer and session management system, enabling fast access to frequently used data and improving platform responsiveness [17].

**e) Cloud Service:** To ensure scalability, reliability, and global accessibility, VCaiLuong utilizes the following cloud services:

**Azure Cloud Services:** Microsoft Azure provides a suite of cloud computing services for hosting, scaling, and managing VCaiLuong’s infrastructure, ensuring high availability and performance for global users [18].

**Firebase:** A platform by Google, Firebase offers tools for authentication, real-time database management, and hosting, enhancing VCaiLuong’s ability to deliver dynamic content and seamless user experiences [19].

## 3.2 System Architecture

This section provides a detailed description of the **VCaiLuong system Architecture** within the context of the overall system design. It includes both a diagram and a thorough explanation of the components, their interactions, and how they contribute to the scalability, performance, and functionality of the system.

### 3.2.1 VCaiLuong Design Overview

The **VCaiLuong system** is designed to offer a scalable and maintainable platform for exploring the Cai Luong cultural heritage and other traditional art forms. The system is structured around two sub-applications: **Visitor Application** and **Admin Portal Application**. These two sub-applications are designed to work independently in terms of source code to ensure modularity, clarity, and ease of maintenance. The system architecture follows a client-server model, where the frontend and backend are split into distinct sites. The frontend applications connect with the backend through APIs, with an **API Gateway** acting as a mediator to ensure that client requests are correctly routed to the suitable backend services (Fig. 3.1).

In response to the potential for high traffic and large requests, especially given the growing interest in traditional art forms, the **VCaiLuong system** is designed using a **microservice pattern** [20]. This approach allows the backend services to scale and perform optimally under heavy loads, ensuring that each service is dedicated to a specific business capability. The VCaiLuong backend is designed using the microservice architecture pattern, which divides the backend into small, independently deployable services. Each service corresponds to a specific business function and is divided into three main components: **VCaiLuong Service or Visitor Service**, which manages accessibility and requests from users exploring cultural content and art forms; the **Admin Portal Service**, which handles requests from the administrative interface for managing content, users, and system configurations; and the **Core Service**, which acts as a central logic layer, managing shared and background tasks for both the **Visitor** and **Admin Portal Services**.

These three backend services work as independent services within the overall microservice ecosystem. They communicate with each other via **Redis Pub/Sub** [21], a message broker pattern, and each backend service is responsible for processing client requests or inter-service requests. This distributed approach allows the **VCaiLuong system** to handle a high volume of traffic while keeping the application components decoupled and easier to maintain. To ensure that the backend services can efficiently handle large numbers of client requests while maintaining high levels of maintainability and scalability, **Domain-Driven Design (DDD)** is applied within each microservice [22]. DDD is a design approach that focuses on aligning the software design with the business domain, making it easier to build and maintain systems that reflect the real-world complexity of the business.

In this architecture, the backend services are divided into modules, where each module handles a specific subset of the application’s functionality. This modular approach ensures that the system’s business logic is clearly separated, making it easier to maintain, extend, and test. Each microservice is structured according to a layered architecture, which is split into three key layers to ensure that responsibilities are clearly defined and managed. These layers include the **Controller (Presentation Layer)**, the **Service (Application Layer)**, and the **Entity and Repository (Domain Layer)**.

The adoption of **microservices** and **Domain-Driven Design** in the **VCaiLuong system** provides several key benefits. First, it enables scalability, allowing the system to scale independently by deploying additional instances of individual microservices based on the demand for specific features (e.g., visitor traffic or administrative tasks). Second, it improves maintainability by splitting the system into smaller services and modules, each with a well-defined responsibility, making the codebase easier to maintain, test, and extend. Third, it provides flexibility as each microservice can be developed and deployed independently, allowing the **VCaiLuong system** to quickly adapt to changes in business

requirements or user needs. Lastly, it enforces a separation of concerns, ensuring that business logic is decoupled from presentation logic, which simplifies the management of each aspect of the application independently.

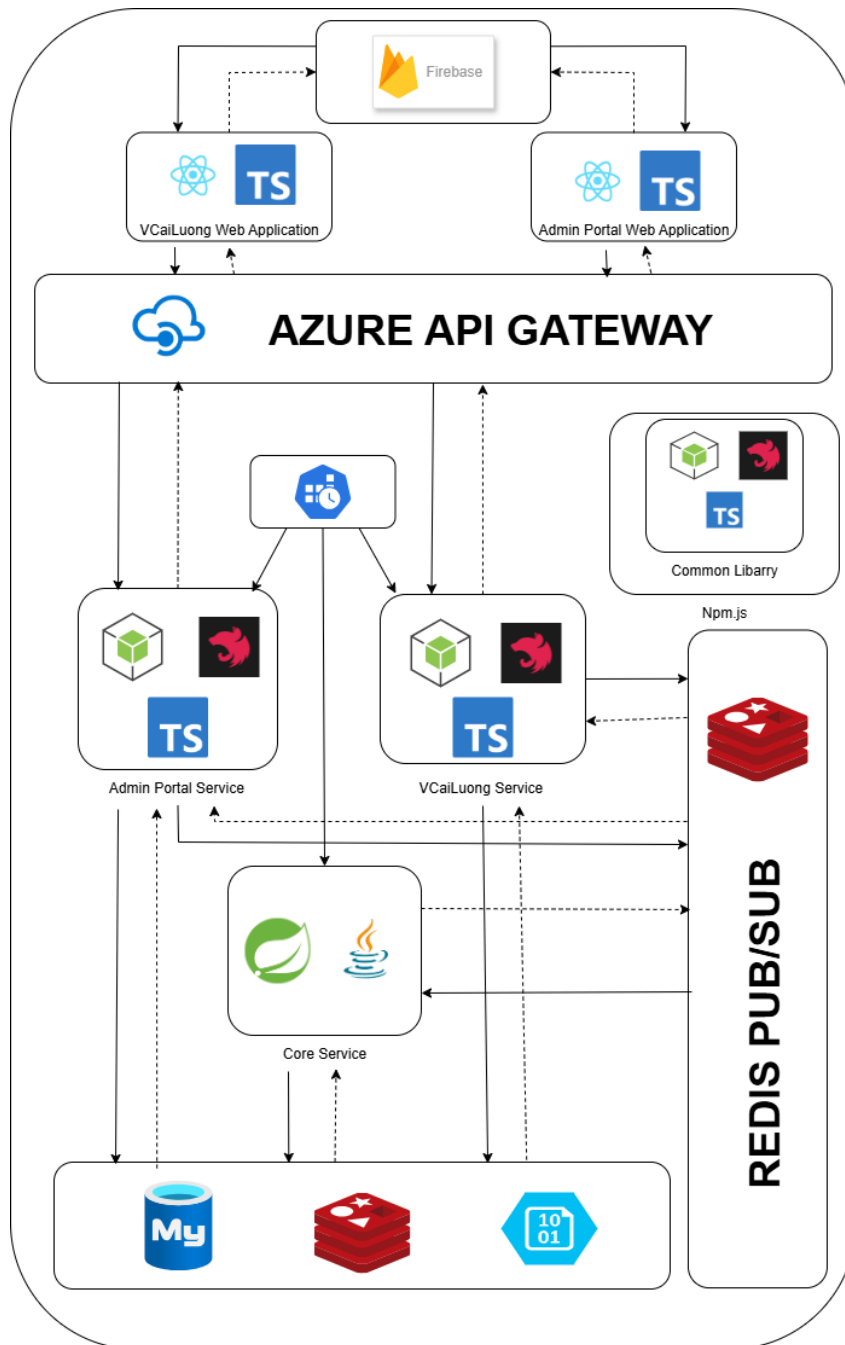


Figure 3.1: VCaiLuong system Architecture Diagram.

### 3.2.2 System Components

The **VCaiLuong system** is designed with a modular and scalable architecture to support its diverse functionality. The architecture is composed of several key components that interact seamlessly to deliver a rich user experience for exploring the Cai Luong cultural heritage and other traditional art forms. Below is a detailed breakdown of the primary components that make up the **VCaiLuong system**.

#### a) Frontend Components of the VCaiLuong system:

The **VCaiLuong system** is structured with two primary frontend applications: the **VCaiLuong Visitor Application** and the **Admin Portal Application**. These applications function as the user interfaces (UI) for the entire microservice ecosystem that powers the VCaiLuong platform. Each application is dedicated to different logic fields, reflecting their distinct purposes within the system. This separation not only simplifies the development process but also ensures that the system is easily maintainable and scalable to accommodate future growth in user traffic or functionality.

The **VCaiLuong Visitor Application** primarily focuses on features that help promote and preserve the **Cai Luong** cultural heritage and other traditional art forms. Its role within the system is to provide visitors with an interactive and immersive experience, encouraging them to explore the rich cultural heritage of Cai Luong through various digital media formats. The visitor application will offer a virtual museum where users can interact with a wide range of artifacts, including pictures, videos, audio recordings, and even 3D models. Users will be able to navigate the collection of artifacts, view detailed information about them, and engage in interactive experiences such as chat boxes and watch parties. These features not only enhance user engagement but also create a dynamic environment where users can communicate with one another while enjoying the cultural content. Through these interactive features, the visitor application serves as a gateway to experiencing the artistic and historical richness of the Cai Luong tradition and other cultural artifacts.

The **Admin Portal Application** is designed as a tool for future development, aimed at museum administrators and content providers. It serves as a comprehensive backend interface for managing the operations of the virtual museum. The portal enables administrators to manage museum-related data, including the creation of new museums, updating information for existing museums, and organizing digital content. In addition, the Admin Portal facilitates monitoring of user activities within the system, such as tracking **watch parties**, which allow users to engage with cultural content collectively in real time. Other key functionalities include verifying and approving new museum templates, managing content categories to ensure diversity within the VCaiLuong system, and overseeing user roles and permissions to maintain appropriate access control and system balance.

#### b) **API Gateway:**

In the **VCaiLuong system**, the **API Gateway** plays a crucial role in efficiently managing the communication between the frontend applications (the **Visitor Application** and the **Admin Portal**) and the various backend services. The **VCaiLuong platform** is designed with a microservice architecture, where different services handle specific business logic, such as managing visitor interactions, administering content, and handling user analytics. The **API Gateway** acts as the **single entry point** for all requests coming from the frontend applications, routing them to the appropriate microservices based on the type of request. This simplifies the **client-side** interaction by abstracting the complexity of communicating with multiple backend services, allowing both visitor and admin applications to access the required resources through one unified endpoint [23].

In addition to routing requests, the API Gateway in the **VCaiLuong system** performs several critical functions to ensure scalability, security, and performance. It manages load balancing by distributing incoming requests evenly across multiple microservice instances, thereby enhancing system availability and fault tolerance. The API Gateway further strengthens security by filtering requests based on whitelisted and blacklisted URLs, preventing unauthorized access to system resources. By centralizing these functions into a single point of control, the API Gateway substantially improves the performance, security, and scalability of the **VCaiLuong system**, enabling it to accommodate increasing user

demands and support the integration of new features in the future.

#### c) **Common Library:**

In the **VCaiLuong system**, several functions are used across multiple services with the same purpose, such as **pub/sub** (publish/subscribe) features or caching functionalities. To ensure that these common functions can be reused efficiently and to avoid unnecessary code duplication, a **Common Library** is created as a custom library, published on **npmjs** [24]. This custom library can be installed and used across various services in the system, enabling them to leverage shared features without the need to duplicate code. This approach not only improves the maintainability and consistency of the system but also reduces the risk of errors that might arise from having multiple implementations of the same functionality in different services.

Furthermore, the **Common Library** is designed to manage the **versioning** of these shared features. This allows the system to handle updates to the library while ensuring that services can roll back to a suitable version if needed. This version control ensures that any changes or improvements made to the shared features in the library do not break existing functionality in the services using it. By centralizing common features into one library, the **VCaiLuong system** can maintain a high level of consistency across all services, streamline development processes, and ensure that code is easily maintained and updated without redundant work.

Importantly, this library represents a technical contribution to the broader development community. Being publicly available, it can be downloaded and utilized by developers who wish to explore or implement new projects based on a microservices architecture. By providing a reusable, well-structured set of features, the library serves as a foundation for accelerating development, promoting best practices, and supporting researchers or practitioners in adopting microservice-based approaches in their own applications.

#### d) **Backend Service Components:**

In the **VCaiLuong system**, three primary **backend services** play vital roles in handling **client requests** and supporting each other through **inter-service communication**. These services include the **VCaiLuong Service**, the **Admin Portal Service**, and the **Core Service**. These services not only manage requests from the frontend but also interact with each other via **Redis pub/sub (publish/subscribe)**, a **messaging system** that enables efficient communication and **functionality sharing** between services.

The **VCaiLuong Service** is responsible for handling all requests from the **Visitor Application**. In addition to managing the **visitor-specific features**, this service also acts as a **proxy** for requests that require the use of **shared functions**. If a request needs a common function, such as **caching** or notification handling, the **VCaiLuong Service** will **publish** the request to **Redis**, where the **Core Service** is responsible for executing the general function. The **Core Service** handles the **background logic**, ensuring that common features are executed and maintained consistently across the system. However, when it comes to requests that are related specifically to visitor-facing features, the **VCaiLuong Service** will route those requests to the appropriate modules with the suitable **business logic** to handle them.

The **Admin Portal Service** manages all requests from the Admin Application. Like the **VCaiLuong Service**, it can function both as a **logic server** and as a **proxy server** when needed. By leveraging **Redis pub/sub**, the **Admin Portal Service** can reuse **common functions** provided by the **Core Service** or other services to execute specific client requests. This allows the **Admin Portal Service** to stay lightweight and modular, only implementing the functionality necessary for the **administration of the platform**,

while still being able to tap into **shared services** when needed.

The **Core Service** is a central component of the VCaiLuong system, handling all **general functions** and **background logic**. This service operates as a background service, executing scheduled tasks and background jobs, such as data processing, notifications, or periodic updates. It is also designed to enable the reuse of general logic functions that can be called by proxy services, like the **VCaiLuong Service** or **Admin Portal Service**, through **Redis pub/sub**. As the centralized service for executing common business logic, the Core Service ensures that many of the requests from the visitor or admin applications go through it, making it essential for maintaining consistency and reducing duplication of logic across the system.

This architecture ensures that the **VCaiLuong system** can handle a variety of client requests while maintaining a high degree of modularity and reusability. By leveraging the **Redis pub/sub mechanism**, the system is able to decouple the services from each other, allowing each service to focus on its specific responsibility while still enabling shared functionality where necessary. This modular approach allows for easy scaling, better performance, and reduced complexity in maintaining the system over time.

#### d) Infrastructure Components:

The **VCaiLuong system** relies on three essential resources: **Database**, **Redis Cache**, and **Blob Storage**, each serving a specific role in the system’s functionality. The **Database** acts as the central repository for storing all application data, including user information, artifacts, museum content, and visitor interactions. It handles queries and returns data to the backend services, ensuring reliable and persistent storage of critical information.

The **Redis Cache** is used to optimize the database by storing frequently accessed data in memory, which allows for faster retrieval than querying the database. It is especially useful for caching static or rarely changing data, such as system settings or temporary codes (e.g., for password changes). By reducing the load on the database, the cache enhances the overall performance and speed of the system [25].

**Blob Storage** is used for managing and storing large multimedia files, such as images, videos, and 3D models. This includes assets like museum templates, background images, and artifacts that are displayed in the virtual museum. Blob Storage enables efficient handling of these files, allowing for fast access and rendering of media-rich content, which is crucial for the 3D museum experience [26].

Together, these resources provide the backbone for the **VCaiLuong system**, ensuring reliable data storage, optimized performance, and the ability to manage large multimedia files effectively. This architecture helps the system scale and operate efficiently while supporting a rich, interactive experience for users and administrators alike.

## 3.3 Database Design

### 3.3.1 Database Overview

The database system in the **VCaiLuong system** is constructed using **MySQL**, a widely used and reliable relational database management system (RDBMS), to efficiently manage data related to training programs, courses, classes, exam groups, and students. MySQL uses a **relational model**, where data is structured in tables with rows and columns, and relationships between different data entities are established using foreign keys [16]. This approach ensures data consistency and integrity, as it supports structured query language (SQL) for managing and manipulating relational data efficiently. The use of MySQL



provides robust querying capabilities, allowing for complex queries and transactions to be performed with ease, which is essential for handling the structured data of training programs and student management.

MySQL's relational model also supports normalization, which helps eliminate redundancy and ensures data integrity by organizing data into tables based on logical relationships. This makes it easier to manage and update the data, ensuring that the system remains efficient and scalable as it grows.

### 3.3.2 Database Concept

In the **VCaiLuong system**, several entities represent the core data structures used across the application. These entities ensure the proper organization and management of data in the system, helping to efficiently handle the various functionalities and requirements. Below are the key entities used in the system:

**a) User Entity:** This entity stores all information about the base user, including application-related data and contact information. It includes fields such as login name, hashed password, email, phone number, and other relevant contact details. This entity ensures the management of user accounts and authentication data.

**b) Role Entity:** This entity represents the roles in the VCaiLuong system, each having different permissions and limitations in the application. For example, a Visitor role would have restrictions, such as not being allowed to create new museums, while an Admin role would have full control over museum creation and management.

**c) Permission Entity:** The Permission Entity defines specific actions or privileges within the VCaiLuong application that can be assigned to different roles. For example, permissions could include actions like viewing a museum, editing museum details, viewing a party, or editing party settings. These permissions help control access to specific features based on the user's role.

**d) Permission Function Entity:** The **Permission Function Entity** groups related permissions in the VCaiLuong application, defining a clear scope for access control. This grouping simplifies management by allowing administrators to assign or revoke permissions at a functional level, ensuring consistent role-based access across the system.

**e) Visit Booking Entity:** This entity stores data related to a user's booking for a private museum visit. It contains information about the specific booking, such as the date and time of the visit, user details, and any associated museum information.

**f) Notification Setup Entity:** This entity represents the scheduling configuration of user notifications for specific museums. It stores information about the type of notification, the last trigger time, and rules to ensure that notifications are not duplicated, following a set schedule for triggering.

**g) Notification Setting Entity:** Supporting the notification setup, this entity stores details about how notifications should be generated. It includes information such as the method of notification (email, SMS, etc.), the specific days the notification should be triggered, and the contact details (email, phone) of the user.

**h) Museum Entity:** This entity represents a specific virtual museum within the VCaiLuong system. It stores key details such as the museum's category, name, description, cover image, start time, end time, and keys that link to additional related data from other entities.

**i) Museum Category Entity:** Museums in the VCaiLuong system are categorized into various types to enhance user browsing and discovery. This entity organizes museums into categories, allowing users to easily search and explore different types of museums,

such as "Historical" or "Artistic."

**j) Museum Template Entity:** This entity defines the template used for a specific museum category. It allows for flexibility in the design and layout of the virtual museum. It includes information about the 3D template file, which is used to create the structure of the museum, such as walls, floors, picture frames, and doors.

**k) Museum Section Entity:** Within a museum template, there may be various objects such as pictures, videos, and audio files. The Museum Section Entity organizes these objects into specific sections to improve the performance of the system when loading and rendering the museum on the frontend. It stores information about the section's name, order, and which objects belong to each section.

**l) Museum Layout Entity:** To ensure that templates can be reused across multiple museums, the Museum Layout Entity represents placeholders for artifacts within each section of the museum. For example, if a section contains 6 artifact objects, there will be 6 corresponding layouts. This entity helps streamline the creation of new museums based on existing templates and ensures consistency in how artifacts are placed in sections.

**m) Museum Object Entity:** This entity represents the data for a specific artifact displayed in a museum. It includes a description, name, and a URL pointing to the specific file (image, video, etc.) stored in cloud storage.

**n) Museum Protocol Tag Entity:** This entity is used to tag specific categories. For example, the CaiLuong category might have tags like "Music" or "Documentation," which are used to display additional information or group similar types of content within the category.

**o) Process Session Entity:** The **Process Session Entity** provides a schema for tracking and executing individual tasks, such as generating 3D images in the Admin Portal Application. It manages the task's state and updates the necessary data throughout the execution process, ensuring that each task can be properly initiated, monitored, and completed.

**p) Author Entity:** The Author Entity represents the creator of a museum. By default, the museum's provider is considered the author. However, in the future, if users post documents through the portal, they can be considered contributors or authors.

**q) Watch Party Entity:** This entity represents a virtual watch party room, containing all relevant information about the room, such as the video file URL, start time, name, room cover image, and duration. This entity supports the functionality for multiple users to watch a museum or video together in real-time.

**r) Application Configs Entity:** This entity stores the configuration settings for the VCaiLuong system, such as email templates, email content, and other configurable data. By using this entity, administrators can modify settings without needing to change the source code, making it easier to update system configurations and adapt to new requirements.

**s) User Event Entity:** This entity stores the data about user action in the VCaiLuong system, which is used to track user interaction in the system with the event creation time.

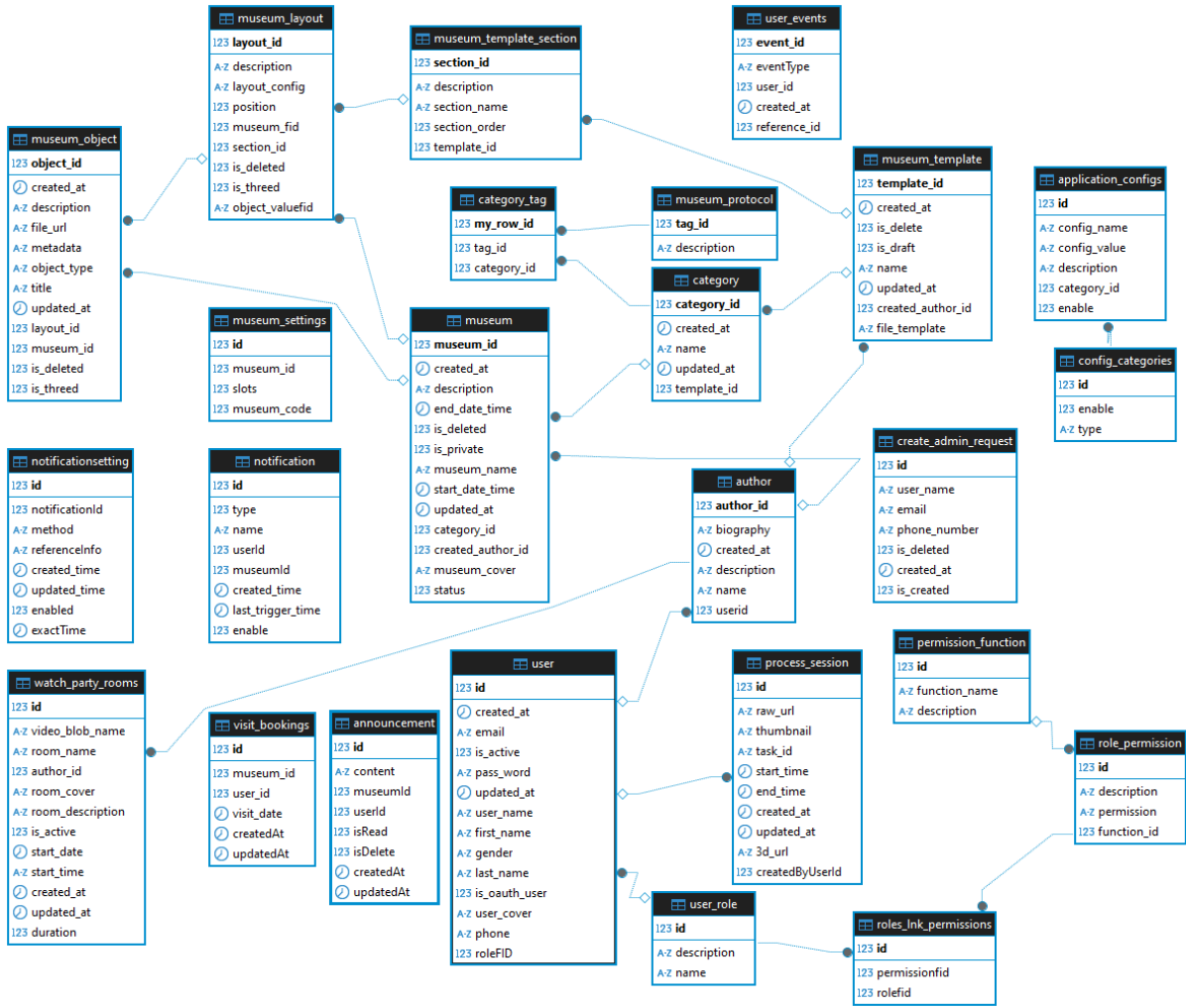


Figure 3.2: VCaiLuong's Database Schema Diagram.

### 3.3.3 Relationship Between Entities

The relationships between the components of the **VCaiLuong system** reflect the logical connections that support the management and analysis of data within the database. These relationships ensure consistency, accuracy, and relevance, allowing the educational institution to evaluate and improve its programs effectively.

**a) User, Role and Permissions:** The relationship between User and Role is characterized by a one-to-one relationship, where each user is assigned a single role within the VCaiLuong system. This structure ensures that each user can only perform a specific set of actions based on their assigned role. Additionally, within each role, there are many Permissions related to that role, and each Permission can be assigned to multiple roles, forming a many-to-many relationship between Role and Permission. By assigning specific permissions to a role, the system can ensure that each role has the appropriate level of access and action privileges within the system.

**b) User, Notifications, Museum:** The relationship between User and Notifications is a one-to-many relationship, where each user can have multiple notification setups. Each notification setup is related to one specific setting, forming a one-to-one relationship between the setup and the settings. Additionally, Notifications are tied to specific Museums, creating a many-to-one relationship between Notifications and Museums. This structure enables users to create and manage multiple notifications for various museums, with each

notification having a specific setting to avoid duplicate triggers.

**c) User, Bookings, Museum:** Users are allowed to book multiple slots for an upcoming private museum visit. This results in a one-to-many relationship between User and Bookings, where a user can book many slots. Similarly, a Museum can have many bookings associated with it, creating a one-to-many relationship between the Museum and Bookings.

**d) Museum, Museum Category:** The connection between Museum and Museum Category is a many-to-one relationship, meaning many museums can be categorized under the same group. This structure helps categorize museums into specific groups with similar attributes, making it easier to organize and find related museums.

**e) Museum Category, Museum Protocol Tag:** Inside a Museum Category, there can be multiple tags used to clarify the content and information about the category. This results in a many-to-many relationship between Museum Category and Museum Protocol Tag, where many tags can be associated with one category, and a tag can be linked to multiple categories.

**f) Museum Category, Museum Template:** The Museum Category and Museum Template have a many-to-one relationship, where multiple museums, across different categories, can use the same template. This structure allows for the reuse of templates across museums, helping maintain consistency and simplifying the creation of new museums with the same visual structure.

**g) Museum Template, Museum Section:** To enhance performance when loading artifacts and to better manage the structure of the museum, each Museum Template contains multiple Museum Sections. This creates a one-to-many relationship between Museum Template and Museum Section, allowing for the division of the museum into smaller sections, which helps in organizing and rendering the museum efficiently.

**h) Museum Section, Museum Layout:** Within each Museum Section, there are multiple Museum Layouts, which act as placeholders for artifacts. This results in a one-to-many relationship between Museum Section and Museum Layout, creating a scalable format to generate new museums with varying artifact layouts within sections.

**i) Museum Layout, Museum Object:** Each Museum Layout contains a Museum Object, which represents a specific artifact placed in the layout. This relationship is a one-to-one relationship, where each layout has a corresponding object (artifact) that is displayed in that layout.

**j) Museum, Author:** A Museum can be linked to one Author, establishing a many-to-one relationship between Museum and Author. By default, the Author is the museum's provider, but in the future, it can include other contributors or creators who add content to the Museum

## 3.4 Integration and Scalability

### 3.4.1 Cloud Service Overview

The **VCaiLuong system** is designed to be a scalable and reliable platform, providing users with a dedicated space to explore the rich Cai Luong culture and other traditional art forms. At the core of the system is a virtual museum, where digitalized artifacts are stored in large-scale storage solutions. These artifacts, ranging from images and videos to 3D models, are easily accessible, ensuring high performance and a seamless user experience. The architecture of the system is built to support scalability and flexibility, allowing for an efficient way to manage and render large amounts of multimedia content.

To ensure the system can scale effectively and maintain data integrity, the **VCaiLuong system** leverages cloud computing services for integration and performance optimization. The system is split into smaller, independent microservices, each responsible for a specific task. These microservices need to communicate with each other efficiently, and cloud services provide the necessary infrastructure to connect and manage these services. By utilizing cloud solutions, the system can handle high traffic, perform background jobs, and process large volumes of data without compromising on performance.

The **VCaiLuong system** primarily relies on two cloud services: **Microsoft Azure** and **Firebase**. **Microsoft Azure** provides a variety of cloud solutions, such as **Blob Storage** for storing large multimedia files and **Azure Database** for managing structured data. Azure’s services ensure that the backend of the system remains robust and scalable [26]. On the other hand, **Firebase** is used for managing **user authentication** functionality. **Firebase** simplifies the process of user authentication by providing an efficient and secure system for handling user sign-ups, logins, and user management. This allows the **VCaiLuong system** to authenticate users across different devices and platforms seamlessly, ensuring a smooth user experience while maintaining security [19]. By integrating **Firebase** for authentication, the system can track user engagement and offer personalized experiences, enhancing interactions with the virtual museum.

### 3.4.2 Integration

The **VCaiLuong system** utilizes a variety of **Microsoft Azure**, **Railway**, **Firebase** services to ensure scalability, high performance, and secure operation.

a) **Azure Database** is used for deploying and managing the system’s database. It efficiently handles high volumes of traffic and queries, ensuring that the database remains responsive while managing large sets of structured data, such as user information, museum content, and other critical system data [27].

b) **Azure Blob Storage** serves as the central storage for the **VCaiLuong system**. It is used to store and retrieve large multimedia files, such as artifacts, 3D models (GLTF files), and essential configuration files like email templates. This enables the system to manage vast amounts of media and static data with high performance and reliability [26].

c) **Azure Redis for Cache** is employed to improve the system’s performance by providing fast, in-memory data caching. Redis supports the Redis Pub/Sub messaging system, which facilitates communication between microservices within the **VCaiLuong ecosystem**. In addition, Redis is utilized for implementing distributed job queues using libraries like BullMQ, and caching static responses to reduce high query execution loads on the database [28].

d) **Azure Gateway** is used to create an API gateway that directs API requests from the frontend to the appropriate microservice within the system. This ensures that each request is routed to the correct service, enabling smooth communication between the frontend and the various backend services of the **VCaiLuong system** [29].

e) **Azure Service App and Railway** are used to deploy the system’s microservices into a production environment. These platforms manage and monitor the services through logging and operational tools, ensuring that each microservice functions efficiently and that the overall system performs reliably in a production setting [30, ?]. f) For secure user authentication, **Firebase OAuth** is integrated into the **VCaiLuong system**. It provides a secure and seamless login experience by using Firebase as the OAuth provider, allowing users to authenticate across multiple platforms with ease and ensuring that sensitive user data is protected [31].

### 3.4.3 Scalability

The **VCaiLuong system** is designed to scale as the project grows, leveraging cloud services to ensure that performance can be enhanced as demand increases. By utilizing services such as **Microsoft Azure** and **Firebase**, the system can scale both horizontally and vertically to handle higher traffic, larger data volumes, and more complex interactions.

Cloud services provide the flexibility to add additional resources such as storage, processing power, and network capacity without requiring significant changes to the underlying architecture. For example, **Azure Blob Storage** can scale to accommodate an increasing number of digital artifacts, while **Azure Redis Cache** can handle growing caching needs and improve response times. Additionally, **Firebase** allows for seamless management of authentication and real-time data processing, which can scale to handle more users and interactions as the system expands.

The efficiency and benefits of integrating these cloud services for scalability are summarized in Table 3.1. This scalability ensures that as the **VCaiLuong system** grows, it can continue to deliver high performance and meet the needs of users without compromising on service quality. Whether handling more user requests, adding new features, or storing additional content, the system can scale efficiently by simply leveraging cloud resources, making it well-suited for future expansion.

Table 3.1: Cloud Service Integration and Scalability Benefits in VCaiLuong

| Cloud Service           | Scalability Type    | Functionality Supported                     | Benefit to VCaiLuong System                                   |
|-------------------------|---------------------|---------------------------------------------|---------------------------------------------------------------|
| Azure Blob Storage      | Vertical/Horizontal | Storage of digital artifacts                | Handles increasing volume of artifacts without downtime       |
| Azure Redis Cache       | Horizontal          | Caching for fast data retrieval             | Improves response time and reduces load on backend services   |
| Azure App Services      | Horizontal/Vertical | Deployment of microservices                 | Supports scaling web and API services dynamically             |
| Firebase Authentication | Horizontal          | User authentication and identity management | Supports more simultaneous users with secure access           |
| Firebase Real-time DB   | Horizontal          | Real-time data processing and syncing       | Ensures smooth handling of increasing interactions and events |

# Chapter 4

## IMPLEMENTATION

This chapter presents an in-depth examination of the implementation of the VCaiLuong Visitor feature. The functionality is built upon a combination of two core components: **user-initiated events**, which require direct interaction from the user, and **automated features**, which are executed independently by the **VCaiLuong system**. Together, these components ensure an engaging, responsive, and seamless experience for users as they explore the virtual Cai Luong museum environment.

### 4.1 Messaging Service Implementation

The **VCaiLuong system** adopts a **microservices architecture**, wherein the backend is decomposed into multiple domain-specific services. Each microservice encapsulates its own business logic and operates independently. However, certain system functionalities require inter-service communication, necessitating a structured and efficient messaging mechanism.

To address this, the VCaiLuong system implements a **messaging service module** using **Redis Pub/Sub**. This messaging mechanism facilitates communication between services in a decoupled and scalable manner. The messaging module is designed as a shared component, stored in a centralized codebase, and is reused across various services within the system.

The messaging service comprises **three primary components**:

- **Proxy Service**: Responsible for publishing requests to a Redis channel when a service requires the functionality of another service.
- **Redis Channel**: Acts as a message queue, temporarily holding messages and enabling asynchronous communication between publishers and subscribers.
- **Consumer Service**: Listens to specific Redis channels, processes incoming messages according to its internal logic, and publishes a response back to a designated response channel.

**The communication flow** follows these steps reference to (Fig. 4.1):

- When a microservice needs to invoke logic from another service, it publishes a request message to a predefined Redis channel via the **Proxy Service**.
- The **Consumer Service**, which is subscribed to the corresponding Redis channel, retrieves the message, executes the required business logic, and then publishes a response message to a dedicated response channel.
- The **Proxy Service**, also subscribed to the response channel, receives the response and completes the request-response cycle, effectively abstracting inter-service communication.

This messaging approach offers several distinct advantages. First, because all traffic is routed through an authenticated, encrypted Redis channel, the attack surface is minimized and overall system security is strengthened. Second, the **pub/sub pattern** keeps services loosely coupled; each microservice can evolve or scale horizontally without requiring intrusive changes in its peers, thereby improving both scalability and long-term maintainability. Finally, by centralizing the messaging logic in a common module, the design

maximizes code reusability and enforces consistent communication practices across the entire **VCaiLuong ecosystem**, eliminating redundancy and simplifying future feature integration.

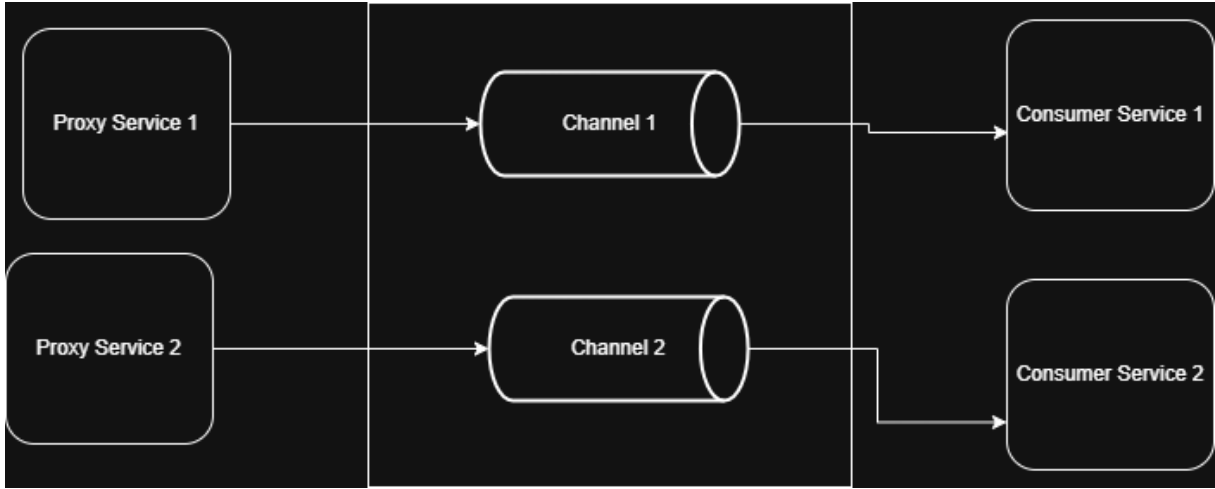


Figure 4.1: Redis Pub/Sub Communication Flow.

## 4.2 Authentication and Authorization Implementation

**VCaiLuong Visitor Application** offers a variety of authenticated access options to ensure secure and personalized entry to the platform’s diverse cultural and educational content. These authentication mechanisms are designed not only to safeguard user data and system resources but also to enhance the overall user experience. The implementation emphasizes both **security**, to protect sensitive resources, and **usability**, enabling seamless access to Cai Luong and other traditional art forms in a trusted digital environment.

### 4.2.1 Basic Authentication Implementation

The **VCaiLuong Visitor Application** provides a familiar and secure authentication mechanism consisting of two key features: **Register** and **Login**. These functions rely on collecting user information to ensure proper account creation and access control.

In the **registration process**, the user provides essential information such as a login username, email address (used for contact and verification), password (stored only after secure hashing), full name, and gender. Once the form is submitted, client-side validation checks the completeness and correctness of the input. The validated request is then sent to the **VCaiLuong server**, where further server-side validation is performed. The server securely hashes sensitive data like the password to protect user credentials. After validation, a confirmation email is generated and sent to the user, containing an activation link. The user must click this link to activate their account before logging in.

#### Steps in the Registration Process:

- User inputs email and password.
- User submits the login form.
- Client-side validation checks the input format.
- Server compares the input password with the stored hashed password.



- If the credentials are invalid, the server returns a **Bad Request** response.
- If the credentials are valid, the server generates **access** and **refresh** tokens.
- Tokens are returned to the client.
- Tokens are stored on the client side (e.g., local storage or global state management) for future authorized requests.

Figure 4.2: Register UI of Visitor Application.

In the **login process**, the user enters their registered email address and password. After submitting the form, client-side validation checks the format and presence of required fields. The server then compares the entered password with the stored hashed password. If the credentials do not match, a Bad Request error is returned to the client. If the credentials are valid, the server generates an access token and a refresh token using a secure secret key. These tokens are returned to the client and stored in local storage or a global state manager. They are used in future authenticated requests, enabling secure access to the system without requiring the user to log in repeatedly.

#### Steps in the Login Process:

- The user enters their email address and password into the login interface.
- The user submits the login form to initiate the authentication process.
- Client-side validation is performed to ensure that the input format (e.g., email structure and password length) meets required criteria.
- The server receives the credentials and compares the provided password against the securely stored hashed password in the database.
- If the credentials are invalid, the server responds with a **400 Bad Request** error, indicating authentication failure.
- If the credentials are valid, the server generates a pair of tokens: an **access token** and a **refresh token**.
- These tokens are returned to the client as part of the successful authentication response.
- On the client side, the tokens are stored securely (e.g., in local storage or a global state management system) for use in future authorized API requests.

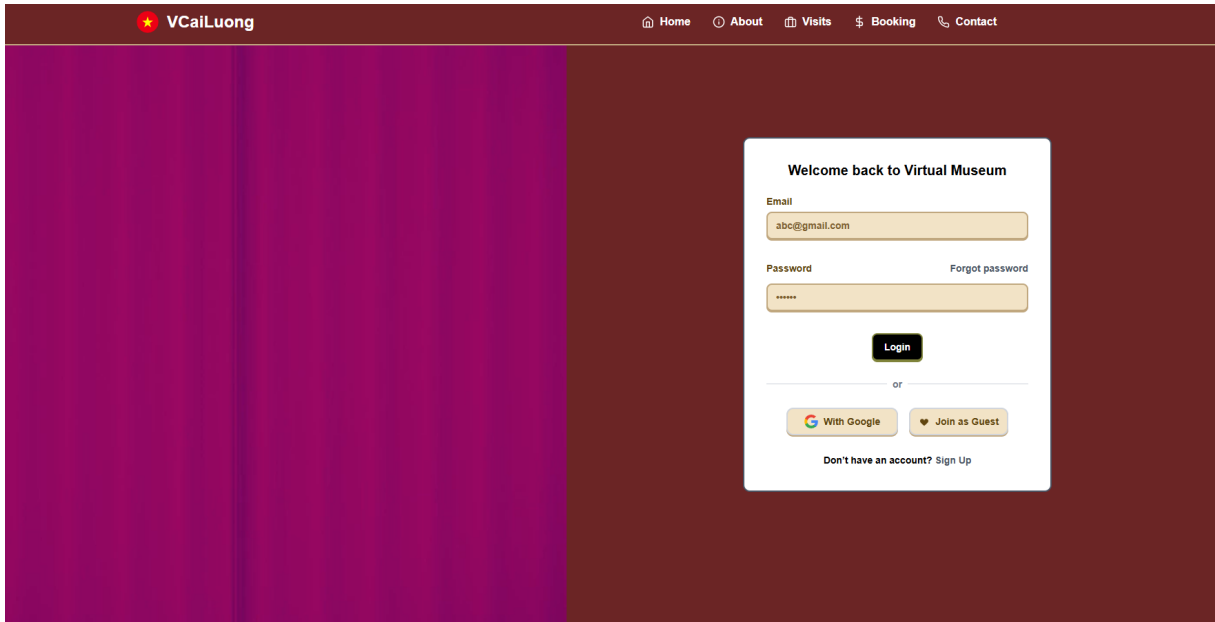


Figure 4.3: Login UI of Visitor Application.

## 4.2.2 OAuth Authentication Implementation

In addition to the standard registration and login methods, the **VCaiLuong Visitor Application** supports authentication through the **OAuth protocol**, which enhances the user experience by enabling secure and streamlined sign-in using third-party providers such as Google.

The OAuth integration is implemented using **Firebase Authentication** as the **OAuth provider**. This setup allows the VCaiLuong system to securely connect with external identity providers to retrieve verified user credentials.

**The OAuth authentication flow proceeds as follows:**

- **OAuth Provider Configuration:** The *VCaiLuong* system is configured to use **Firebase Authentication**, which serves as the intermediary between the application and third-party OAuth providers (e.g., Google).
- **User Interaction:** When the user clicks the “Sign in with Google” button (or similar), the client opens a secure popup window managed by the OAuth provider.
- **Permission Grant:** The user is prompted to grant permission for the *VCaiLuong* application to access their basic profile information from the selected third-party provider.
- **Credential Retrieval:** Upon user approval, Firebase returns an OAuth credential (e.g., access token) to the client.
- **Backend Processing:**
  - The client sends the credential to the *VCaiLuong* backend (Visitor Service).
  - The backend verifies the credentials with the third-party provider and retrieves the user’s basic profile information.
- **Account Handling:**
  - If the user does not already exist in the system, a new user record is created automatically using the retrieved profile data.
  - If the user already exists, the system proceeds to token generation.

- **Token Generation:** The backend generates an **access token** and **refresh token**, following the same process as the standard login flow.
- **Response:** The tokens are returned to the client, where they are securely stored (e.g., in local storage or global state) for use in subsequent authorized requests.

This OAuth flow not only simplifies the user onboarding process but also maintains a high standard of security by offloading credential management to trusted identity providers.

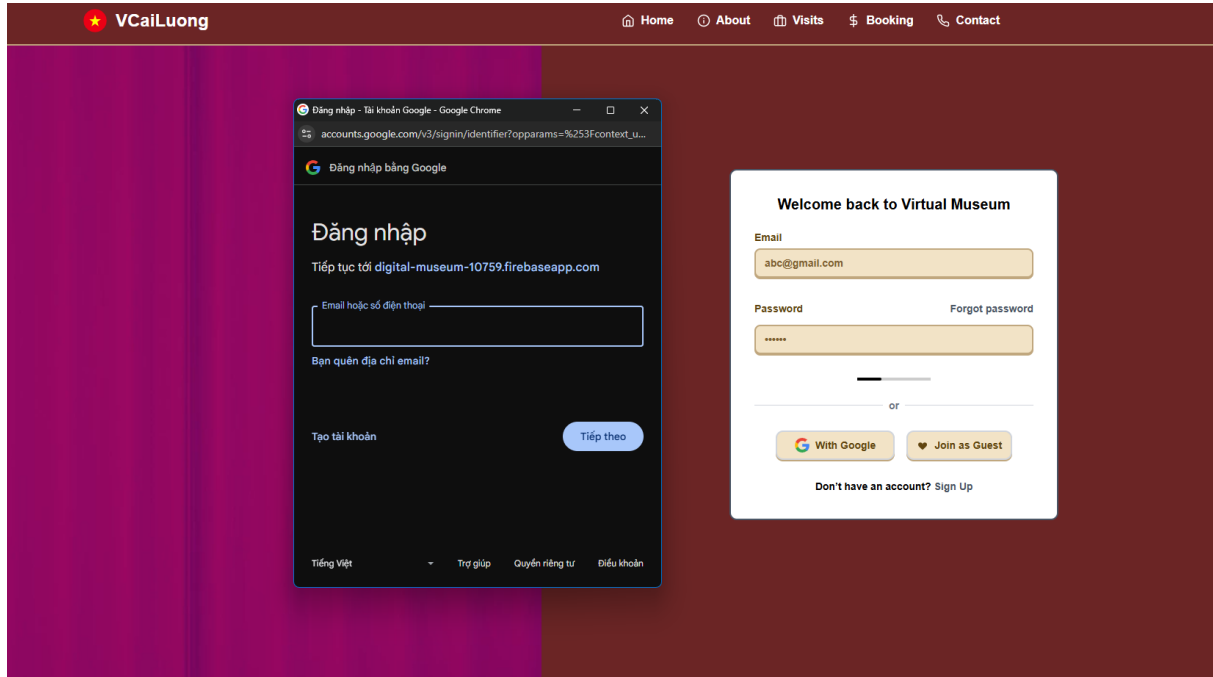


Figure 4.4: OAuth UI of Visitor Application.

### 4.2.3 Authorization Process Implementation

The **VCaiLuong Visitor Application** implements **token-based authorization** to securely manage and verify requests from clients. This approach plays a crucial role in maintaining secure communication between the frontend (client) and backend (server), especially in distributed architectures like microservices.

#### a) Token-Based Authorization and JWT

Token-based authorization is a security mechanism where the user's identity is validated once—typically during login, and a **token** is issued as proof of authentication. This token is then included in the headers of future requests to access protected resources, eliminating the need to repeatedly submit sensitive credentials like a password.

In VCaiLuong, once the user successfully logs in or signs in via OAuth, the server generates a **JWT (JSON Web Token)**. This token encapsulates user-specific claims (e.g., user ID, role, expiration time) and is cryptographically signed to ensure authenticity and integrity.

A JSON Web Token (JWT) is a compact, URL-safe, and stateless format for securely transmitting information as a JSON object. It consists of three parts: the **header**, which specifies the signing algorithm (e.g., HS256); the **payload**, which includes user-related claims such as user ID, email, role, and expiration time; and the **signature**, a hash generated using the header, payload, and a server-side secret key to ensure the token's integrity and authenticity [32].

### b) Authorization in the VCaiLuong Application

Both the **Visitor** and **Admin** applications in the VCaiLuong system adopt **token-based authentication** as the foundational mechanism for securing API communication and user session management. This method significantly enhances overall system security by enforcing strict verification of each incoming request from the client. Rather than relying on traditional session identifiers or persistent login credentials, the system utilizes short-lived **access tokens** and long-lived **refresh tokens**, ensuring that sensitive operations are only permitted for authenticated and authorized users.

When a client initiates an API request to access a protected resource, the access token, previously issued during the login or OAuth authorization process, is included in the HTTP request header. Upon receiving the request, the backend server first parses the access token and attempts to validate it by referencing a **Redis cache**, which acts as a high-speed, in-memory store for previously verified tokens. If the token is located in the cache and remains valid (i.e., has not expired or been revoked), the server can immediately grant access to the requested resource, thereby optimizing performance by avoiding repeated validations.

This caching strategy reduces the computational overhead associated with token validation and enables the system to handle a higher volume of authenticated requests efficiently. Figure ?? illustrates the token verification flow, highlighting the cache lookup as the first step in the decision chain.

If the token is **not found** in the cache or has expired, the server forwards a **proxy request** to the **Core Service** to validate the token.

- If the token is **invalid**, the request is denied, and the client is not granted access to the protected resource.
- If the token is **valid**, the result of the validation is cached with a short **Time-To-Live (TTL)**, enabling faster authentication checks for subsequent requests during that period.

In cases where the **access token** is invalid or expired, the system checks whether the client has a valid **refresh token**. If so, the client can send a request to the server with the refresh token. The server verifies this token:

- If the **refresh token** is valid and not expired, the server issues a new **access token** to the client.
- If the **refresh token** is invalid or has expired, the system denies the request and does not generate a new token.

This mechanism balances performance (through caching), security (via strict token validation), and user experience (by supporting seamless token renewal).

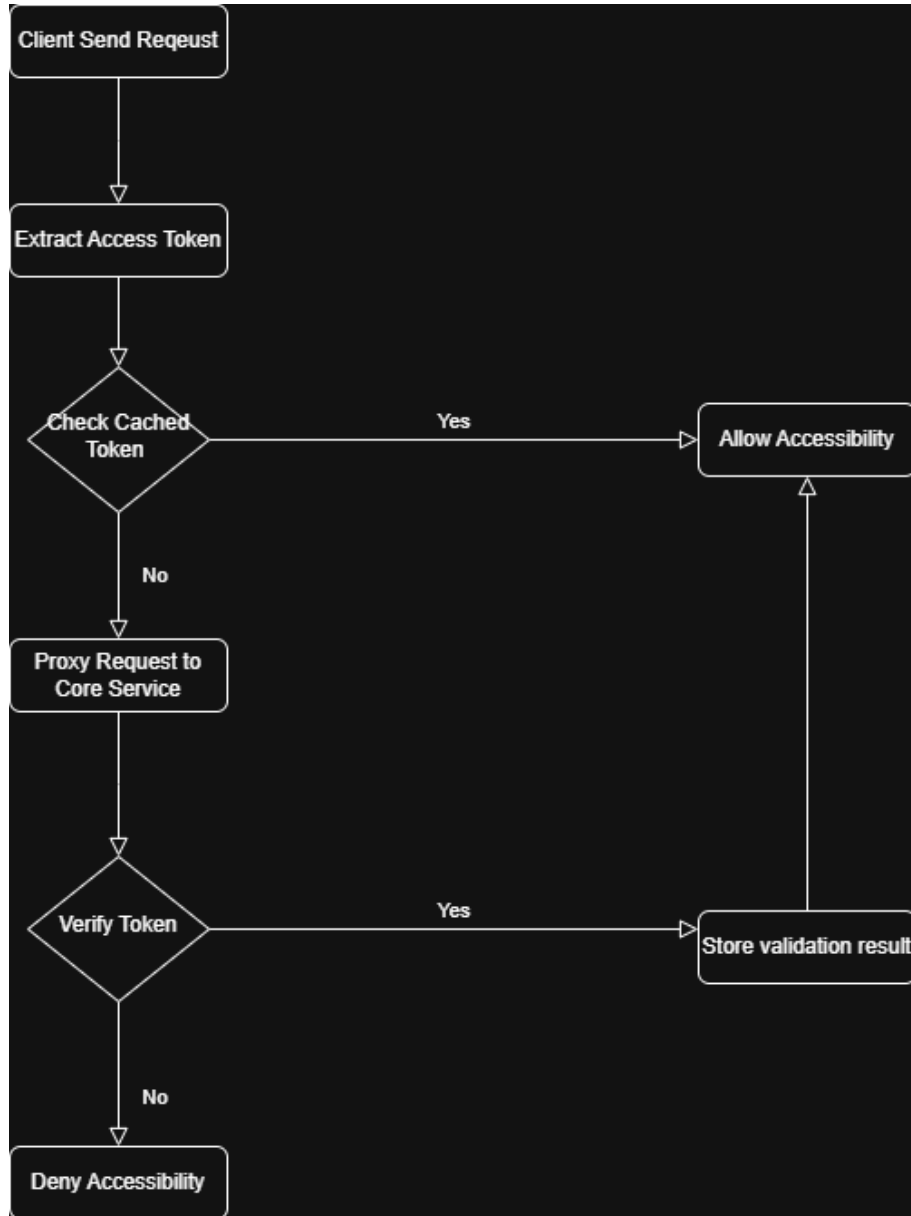


Figure 4.5: Authorization Process of VCaiLuong System.

## 4.3 Notification Features Implementation

The **VCaiLuong Visitor Application** offers users a comprehensive feature to configure personalized notifications for upcoming museum events. This functionality is designed to enhance user engagement by providing timely reminders according to the user's preferred schedule and delivery method.

### 4.3.1 Creating and Editing Implementation

Users are provided with a variety of notification types to suit their preferences and schedules. These include **Once**, which allows users to receive a one-time notification on a specific date; **Daily**, which delivers notifications every day; **Weekly**, which sends reminders once per week; and **Monthly**, which schedules notifications to occur once per month. In addition to selecting the notification frequency, users can also choose from

multiple delivery methods. These methods include **Email**, which sends alerts to the user's registered email address; **SMS**, which delivers messages directly to their mobile device; and **Application Announcement**, which provides in-app notifications through the VCaiLuong Visitor Application. Users may select one or multiple methods based on their preferences to ensure timely and convenient reminders.

**The setup and editing of notifications follow a guided workflow that includes the following actions:**

- **Select Notification Type:**

- The user begins by selecting one of the four available notification types.
- If the user selects the **Once** type, they are required to choose a specific date before proceeding.

- **Choose Notification Method(s):**

- At least one delivery method (Email, SMS, or Application Announcement) must be selected.
- Multiple delivery methods may be selected concurrently.

- **Client-side Validation:**

- Validation is triggered when the user clicks the **Submit** button.
- The application ensures that a valid notification type is selected.
- At least one valid delivery method is chosen.
- If applicable, a specific date is provided for the **Once** notification type.

- **API Communication:**

- Upon successful validation, the client sends the request to the backend via a designated API.
- The API Gateway routes the request to the appropriate internal service.

- **Server-side Processing:**

- The *Visitor Service* serves as the primary handler for notification setup and editing.
- Performs an additional layer of validation to ensure data integrity and consistency.
- Constructs a **Data Transfer Object (DTO)** from the validated input.
- Persists the DTO to the notification scheduling database.

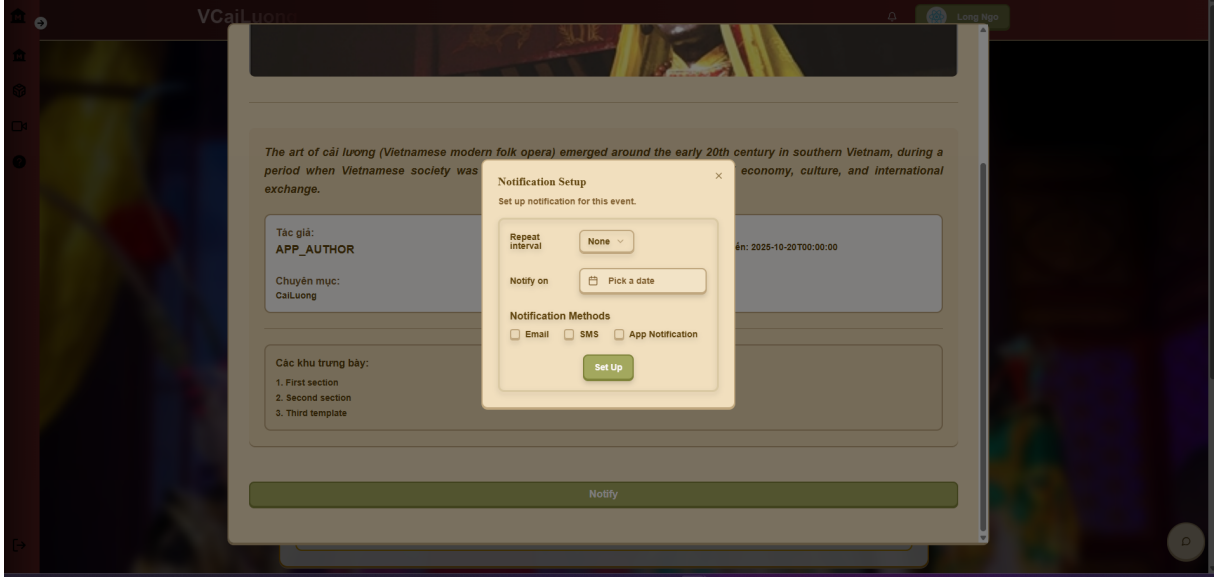


Figure 4.6: UI Design for Creating Notifications in VCaiLuong.

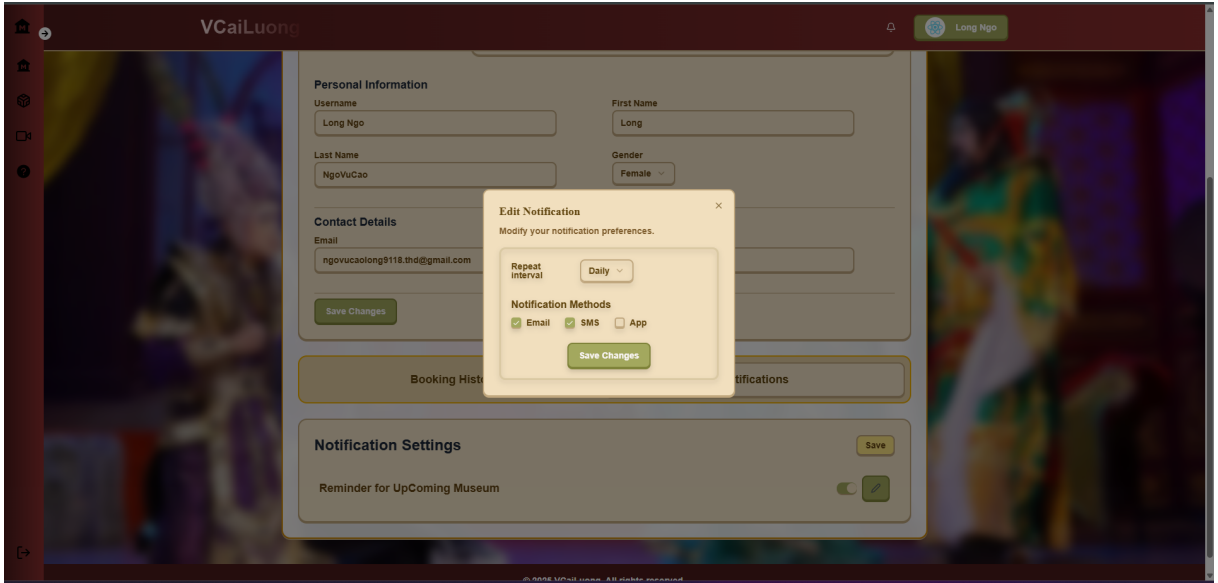


Figure 4.7: UI Design for Creating Notifications in VCaiLuong.

### 4.3.2 Auto-Generate Notification Implementation

The automatic generation of notifications in the **VCaiLuong Visitor Application** is managed by a scheduled cron job that triggers the **Visitor Service** at predefined intervals. A **cron job** is a time-based task scheduler that automates the execution of specific processes, ensuring they run consistently without manual intervention. In this context, it plays a critical role in ensuring that notifications are generated exactly according to each user's configured schedule.

For notification setups' job with a **daily schedule**, the system is designed to execute the generation process every morning, during a period of minimal server load, to optimize performance and system resource usage. This timing enables the backend service to efficiently handle the resource-intensive task of generating notifications, which involves

interactions with several core components, including the **Redis cache**, **database systems**, and support from **Core Services**. By aligning the process with off-peak hours, the system ensures high availability, maintains responsiveness, and guarantees that user notifications are generated and delivered in a timely and reliable manner.

#### a) Job Queue in the Notification Creating Process

In the **VCaiLuong Visitor Application**, the job queue pattern is specifically applied during the notification creation process to ensure scalability and reliability. When the system initiates the generation of notifications, typically triggered by a scheduled **cron job**, it first queries all valid notification setups scheduled for that day. Each setup is then treated as a separate job, which is pushed into a **job queue** managed by **BullMQ**, a **message queue library** that operates on top of Redis [33].

Once the **jobs** are enqueued, **workers** are responsible for fetching and processing them one by one. Each worker handles a single job independently, executing the necessary logic to generate a user notification. This includes tasks such as constructing the notification content, determining the delivery method (email, SMS, or in-app), and saving the result to the database.

To ensure data integrity and consistency, each job is processed within a **database transaction**. If any error occurs during processing, the transaction is **automatically rolled back**, preventing partial or invalid data from being committed. The integration of a **job queue** enables the system to handle a large volume of notifications efficiently, ensures fault isolation, and maintains high system availability.

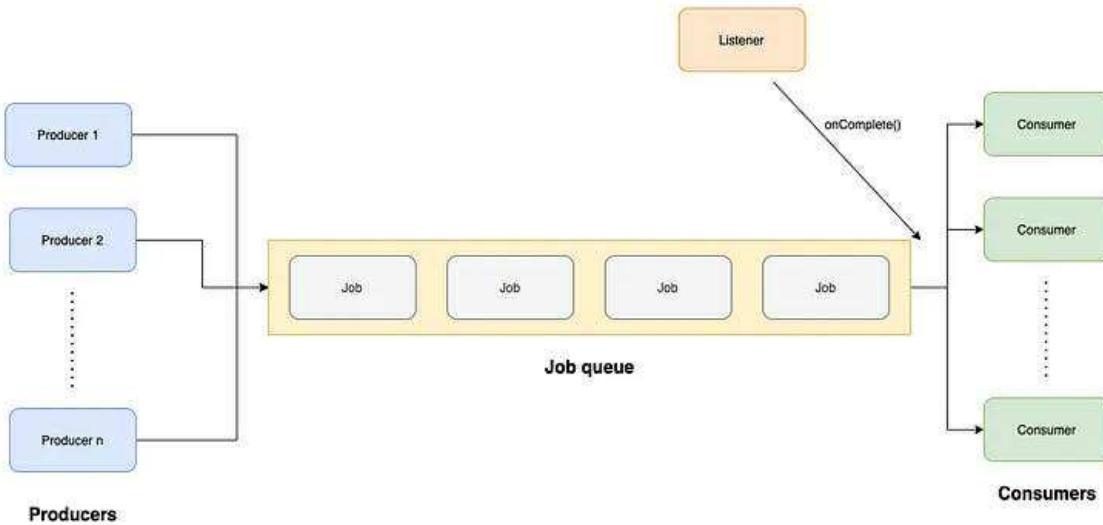


Figure 4.8: Job Queue Structure in the BullMQ Framework.

#### b) Generate Notification Process

The notification generation process in the **VCaiLuong Visitor Application** is designed to run automatically and reliably using a combination of scheduled tasks, job queues, and transactional job handling. The steps below outline the full lifecycle of the process:

- **Cron Job Trigger:** A scheduled cron job is configured to activate the notification generation service at a specific time each day, typically during off-peak hours, to optimize performance.



- **Query and Job Creation:** Upon activation, the *VCaiLuong Visitor Service* queries the system for all valid notification setups scheduled for the current day. For each valid setup, the service creates a job and enqueues it in the job queue. Each job is assigned a unique job ID derived from the setup’s unique identifier to prevent duplicate processing.
- **Transaction Initialization:** When a worker picks up a job, it begins by initializing a new database transaction. This guarantees that all actions within the job maintain **ACID** (Atomicity, Consistency, Isolation, Durability) properties [34], allowing rollback in case of failure.
- **Job Processing Workflow:** The worker processes the job using the notification setup as input, following these sequential steps:
  - **Proxy Communication:** The worker makes a proxy call to the *Core Service* to retrieve necessary base information (e.g., user profiles, event metadata), promoting service reuse and reducing duplication.
  - **Type and Method Validation:** The system verifies the notification type and validates the scheduled day, particularly for notifications of type **Once**, which must match the current date.
  - **Notification Dispatch:** Based on the selected delivery methods (Email, SMS, or Application Announcement), the system sends the notifications through the corresponding channels.
  - **Transaction Commit:** If all operations are executed successfully, the transaction is committed, thereby finalizing the creation and delivery of the notification.

This structured and resilient workflow ensures that notifications are generated accurately, system integrity is maintained, and user configurations are honored consistently.

## 4.4 Booking Feature Implementation

Besides setting up the notification process, **VCaiLuong Visitor Application** also provides create booking slot for upcoming private museums, which have limited visit slots with special artifacts and content that are not disclosed openly.

### 4.4.1 Create Booking Process

The **booking process** in the **VCaiLuong Visitor Application** allows users to reserve a time slot to visit a private museum. This process involves a combination of **user actions** and **system-side operations**, ensuring that bookings are accurately handled and properly confirmed. The system is designed to validate input data, process user requests efficiently, and provide immediate feedback via confirmation emails. The following steps outline the full process from the user’s initial action to final confirmation:

- **Museum Selection:** The user selects a private museum from the list of available options.
- **Display Available Dates:** The system retrieves and displays valid booking dates based on the selected museum’s defined start and end availability.
- **Date Selection:** The user chooses a specific date from the list of available dates.
- **Booking Submission:** The user submits the booking request through the client application.
- **Client-side Validation:** Before submission, the application validates the input to ensure that:

- A valid date has been selected.
- All required fields are properly filled.
- The selected date falls within the allowed booking period.
- **API Request to Backend:** Upon successful validation, the client sends the booking request to the backend services via a designated API.
- **Backend Processing:** The **VCaiLuong Visitor Service** performs the following operations:
  - Receives and verifies the booking request.
  - Processes and stores the booking data in the system’s database.
  - Creates a reminder notification event to automatically send the access code on the selected visit date.
- **Confirmation Email:** After the booking is successfully created, the system automatically sends a confirmation email to the user, summarizing the booking details.

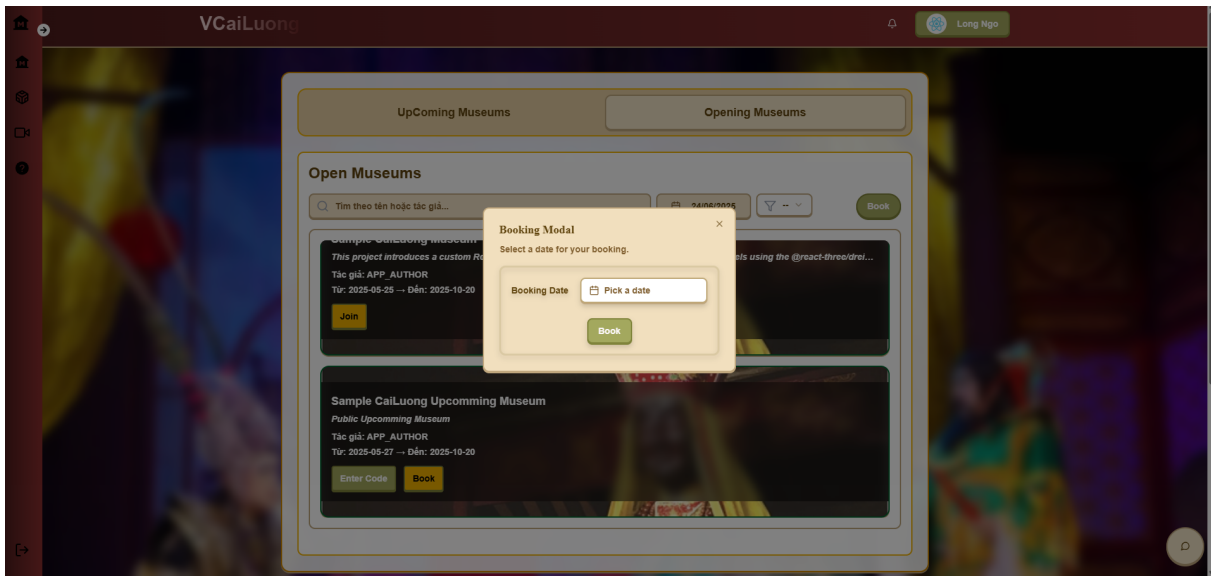


Figure 4.9: UI Design for Creating Bookings in VCaiLuong.

#### 4.4.2 Automatic Access Code Delivery for Bookings Process

To ensure users receive **timely reminders** and **seamless access** to their booked private museum visits, the VCaiLuong Visitor Application includes an **automated feature** for delivering access codes on the scheduled visit date. This functionality is implemented as a **scheduled background job**, which is **triggered daily by a cron job**. Each day, the system scans the booking database to identify entries where the booking date matches the current date. For every valid entry, the system automatically **generates** (if necessary) and **sends the corresponding access code** to the user via the selected communication channels, such as **email** or **in-app notifications**. This daily execution guarantees that no valid booking is missed and that users are consistently provided with their access credentials on time. By automating this process through scheduled tasks, the system enhances reliability, scalability, and overall user experience without requiring manual oversight.

The steps involved in the automatic access code delivery process are as follows:

- **Cron Job Trigger:** A scheduled cron job is executed daily to initiate the access code reminder process.
- **Trigger Notification Generation in Visitor Service:** The cron job triggers the *VCaiLuong Visitor Service*, which begins processing scheduled notifications.
- **Filter Booking Reminders:** The Visitor Service filters all configured notifications and selects those with the type **Reminder**, specifically intended for booking reminders.
- **Create Jobs for Each Booking Reminder:** Each valid booking reminder is treated as a separate job and added to the **Job Queue**, which is implemented using **BullMQ** and **Redis**.
- **Process Jobs with Transaction Handling:** Each job is processed within the context of a database transaction to preserve **ACID** properties. If the job fails (e.g., access code generation or delivery failure), the transaction is rolled back to maintain data integrity.
- **Send Access Code:** During job execution, the system retrieves the appropriate access code from the museum’s daily-generated configuration. This code is refreshed daily to ensure both security and uniqueness. The access code is then sent to the user via the selected notification method, such as **email** or **application announcement**, according to the user’s preferences.

## 4.5 User Information Features Implementation

In the **VCaiLuong Visitor system**, users are granted the ability to view and edit their personal information, which may be either registered manually or retrieved from third-party OAuth providers. This functionality ensures that users can maintain control over their visual identity and contact details, both of which are utilized in key features such as notifications and museum bookings.

Additionally, the **user information module** includes an achievement system, which serves as a motivational element within the application. Through this feature, users can view a collection of their earned achievements, which reflect their engagement with the **VCaiLuong platform**. These achievements function not only as markers of personal exploration, such as visits to museums or participation in watch parties, but also as recognition of the user’s contribution to the preservation and promotion of the **Cai Luong cultural heritage**. As such, the achievement system offers a certificate-like experience, rewarding users for their efforts in discovering and sharing **traditional Vietnamese performing arts**.

### 4.5.1 Viewing and Editing Information Process

The **VCaiLuong Visitor system** implements view and edit functionalities through two sub-processes, separated by data type. One handles textual information, such as username, name, gender, and contact details, while the other manages avatar updates. Text data is processed via backend APIs, whereas avatar files are uploaded to Azure Blob Storage. This design ensures efficient data handling, secure file management, and a streamlined user experience.

#### a) Fetching User Information (Text, File) (Fig. 4.10)

- The client-side application sends a request to fetch user profile data via an API endpoint.

- This API request is routed through the **API Gateway** to the *Visitor Service*, which serves as the central handler for user-related operations.
- The **Visitor Service** performs the following actions:
  - Queries the user's profile information (text fields such as name, email, and gender) from the database.
  - Generates a **Shared Access Signature (SAS) URL** for the user's avatar stored in **Azure Blob Storage**.
  - This SAS URL provides time-limited and permission-restricted access to the avatar file.
- A combined response is sent back to the client, including:
  - User text information (e.g., name, email).
  - Avatar URL (via the generated SAS).
- The client-side application uses this data to display the user's text information and fetch the avatar image securely using the SAS URL.

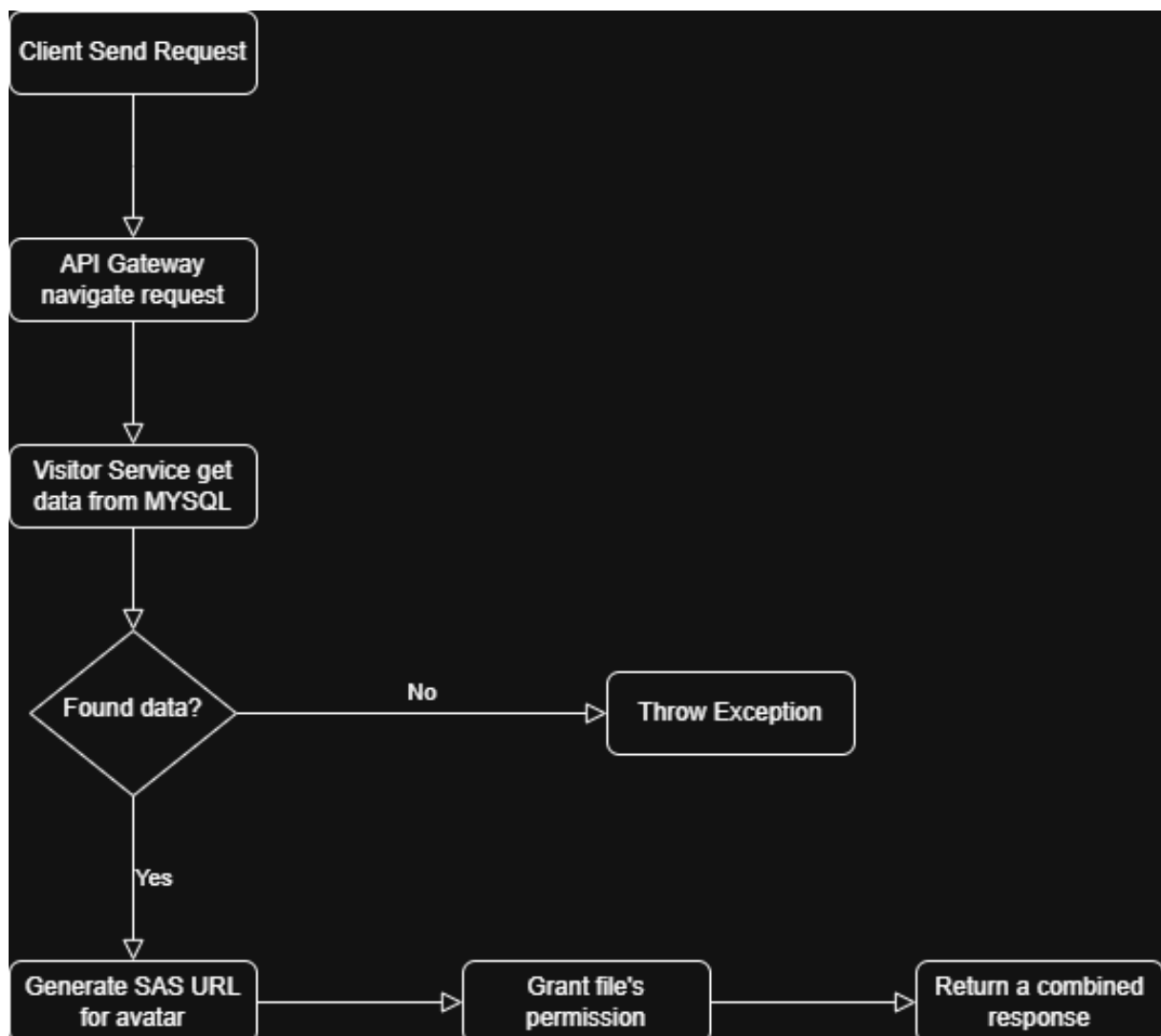


Figure 4.10: Fetching User Information Work Flow Diagram.

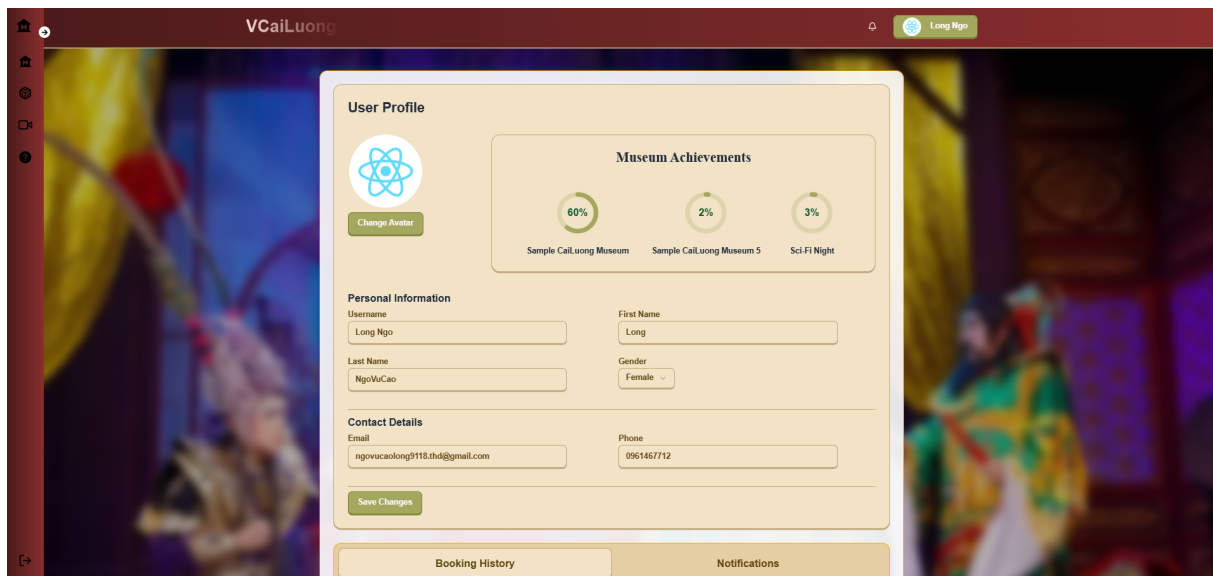


Figure 4.11: User Information Interface in the VCaiLuong Visitor Application.

## b) Editing and Saving User Information Process

### • Saving Text Information

- The user edits text fields (e.g., name, email, gender) on the profile page.
- Upon clicking the **Save** button:
  - \* The client-side sends the updated text information to the backend via an API call.
  - \* The request is routed through the **API Gateway** to the *Visitor Service*.
  - \* The *Visitor Service* updates the user data in the database.
  - \* A confirmation response is returned to the client.

### • Uploading Avatar

- If a user uploads a new avatar:
  - \* The client sends the image file to a dedicated avatar upload API.
  - \* The request is routed through the **API Gateway** to the *Visitor Service*.
- The *Visitor Service* handles the request as follows:
  - \* **Input Validation:** Ensures the uploaded file is a valid image (e.g., checks MIME type and file extension).
  - \* **File Name Generation:** Constructs a unique file name using a combination of user ID and timestamp.
  - \* **Database Update:** Stores the new file name in the user's profile record in the database.
  - \* **File Upload:** Uploads the image file to **Azure Blob Storage** under a structured path, such as `/avatars/{user_id}/`.

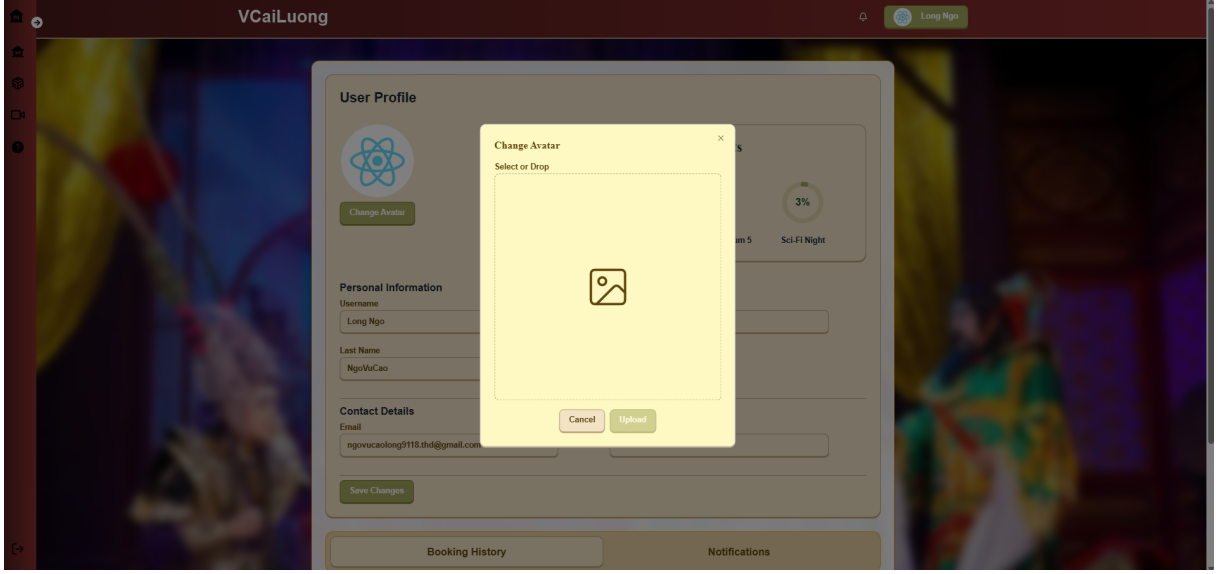


Figure 4.12: Uploading Avatar UI in the VCaiLuong Visitor Application.

#### 4.5.2 Displaying and Calculating Achievement Process

The **VCaiLuong Visitor application** tracks user achievements based on interaction events, such as joining or leaving a museum or watch party session. These actions are logged with timestamps to record interaction duration. The Visitor Service processes this data to calculate user progress using a predefined formula.

$$\text{Progress} = \frac{\sum(\text{Time}_{\text{leave}} - \text{Time}_{\text{join}})}{60} \times 100$$

In this equation, **TimeJoin** and **TimeLeave** represent the timestamps (in minutes) corresponding to when a user enters and exits a session, respectively. The total time spent is summed across all relevant events, converted to hours by dividing by 60, and then scaled by a factor of 100 to express the result as a progress percentage.

The calculated progress value is then attached as an attribute in the "Get User Information" API response. This information is subsequently returned to the client side, where it is displayed to the user as a representation of their engagement or achievement within the application.

### 4.6 Museum Feature Implementation

The **VCaiLuong Visitor application** serves as a digital gateway to experience and explore Cai Luong culture through a virtual, interactive museum. Users take on the role of virtual visitors, granted access to browse exhibits, inspect artefacts, and read descriptive content-closely replicating the feel of a real-world museum. This immersive format appeals especially to younger audiences who are drawn to innovative digital experiences.

The museum feature is implemented through three key components. The first, **Museum Access**, outlines how users enter the virtual space and navigate the system's entry points. The second, **Museum Virtualization**, encompasses the technical foundation on both client and server sides, handling 3D rendering, environment setup, and artifact data management.

The final component, **Implementation of Interactive Features**, focuses on dynamic user interactions within the virtual museum. These include viewing artefact details, accessing multimedia elements, and engaging in guided or self-directed exploration, all designed to deepen cultural engagement and user retention.

### 4.6.1 Museum Accessing Process

The museum access process in the VCaiLuong Visitor application involves a coordinated sequence of steps between the client and server sides, including interface display, condition validation, and access control. This ensures that users are granted appropriate permissions before entering a specific virtual museum room. By supporting multiple access pathways, the system provides users with flexible options to explore Cai Luong culture and other traditional art forms within a rich, interactive environment.

#### Museum List Displaying

For display steps, at initial system will display all accessible museums in the dashboard UI and user can choose to see detailed information for each museum:

##### a) Dashboard Displaying

- **Client Request:** The client side initiates the process by sending an authorized API request to retrieve all accessible museums.
- **API Gateway Routing:** The API Gateway receives the request and routes it to the corresponding service within the *Visitor Services* module.
- **Data Retrieval:** The Visitor Service queries the database and applies filtering conditions based on the opening and upcoming status of each museum. These conditions are evaluated using each museum's configured start and end times. The resulting dataset is returned to the client in the form of a structured response.
- **Client-Side Rendering and Behavior:** Upon receiving the response, the client side renders a list of accessible museums. It also dynamically adjusts available actions based on museum attributes, for example, prompting for an access code if the museum is marked as private.

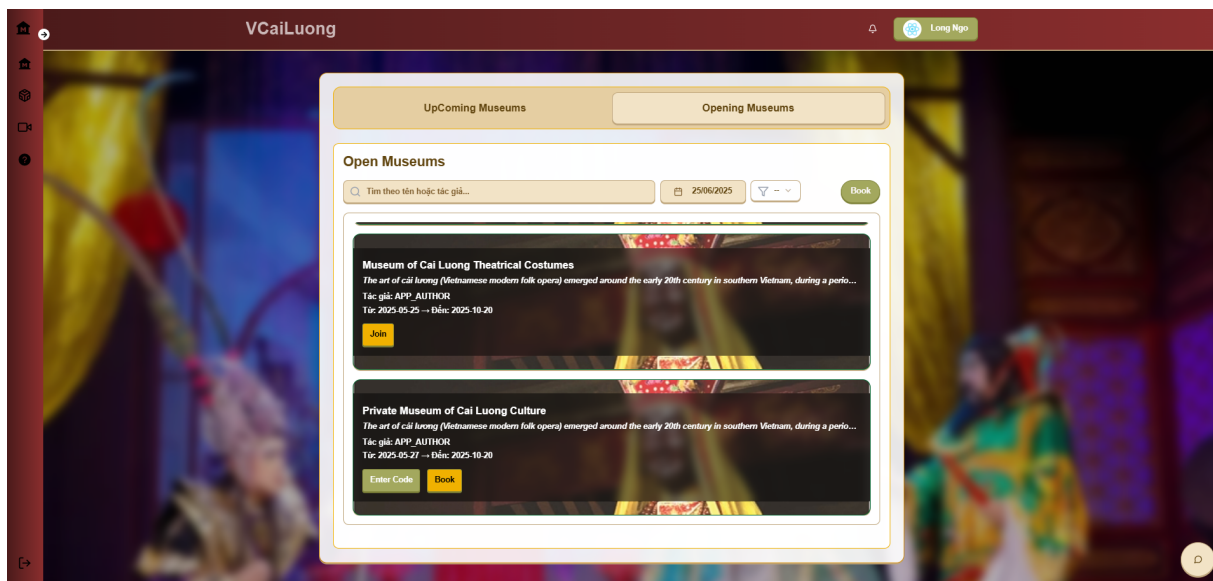


Figure 4.13: Dashboard UI in the VCaiLuong Visitor Application.



## b) Museum Detail Displaying

- **Client Request:** The client side initiates a request to fetch all accessible museums using an authorized API call.
- **API Gateway Routing:** The API Gateway receives and processes the request, routing it to the appropriate service module within the *Visitor Services*.
- **Service Delegation to Core:** The *Visitor Service* delegates the request by making a proxy call to the **Core Service**, which contains the **getMuseumDetail** function. This function is implemented as a shared, reusable component designed to support both the Visitor and Admin portals, thereby promoting modularity and code reuse within the system architecture.
- **Client Response Handling:** Upon receiving the response, the client dynamically renders the user interface based on the returned data, adjusting content and functionality as needed.

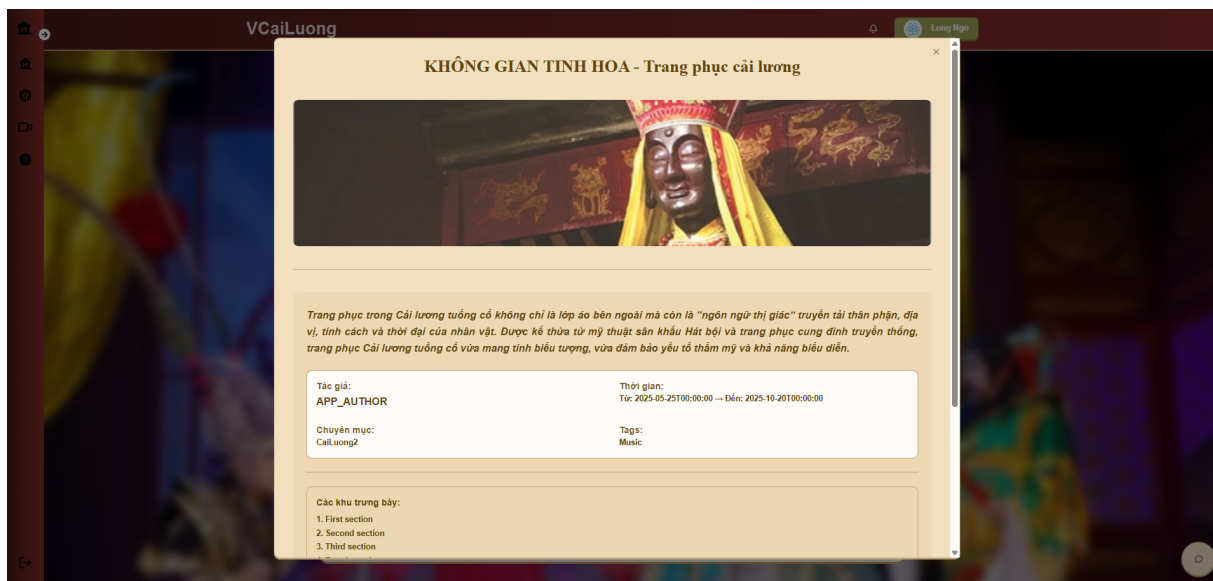


Figure 4.14: Museum Detail UI in the VCaiLuong Visitor Application.

## Museum Accessing

The **VCaiLuong Visitor Application** supports user access to two types of virtual museums: **public** and **private**. This dual-access model enhances flexibility, supports personalized visitor experiences, and accommodates both open exhibitions and curated, limited-access showcases.

- **Public Museums:** These museums are accessible to all users without the need for prior booking. Users can join and explore the museum freely at any time within its defined availability period.
- **Private Museums:** Access to private museums requires users to make an advance booking. An access code is issued, which is only valid on the reserved visit date. This controlled entry system enables museum administrators to effectively manage visitor flow and ensure a more organized experience.

The access workflow involves the following steps:

- **User Input:** The user enters the access code through the designated input field within the application interface.
- **Client Request:** The client sends a request to validate the access code using a dedicated API endpoint.



- **API Gateway Routing:** The API Gateway receives the request and forwards it to the *Visitor Service*.
- **Validation:** The *Visitor Service* retrieves the expected access code for the selected museum from the system's configuration. It then verifies the user's booking and compares the entered code with the expected value.
- **Access Decision:**
  - If the access code matches and the booking is valid, the user is granted permission to enter the private museum room.
  - If the access code is incorrect or the user lacks a valid booking, access is denied.

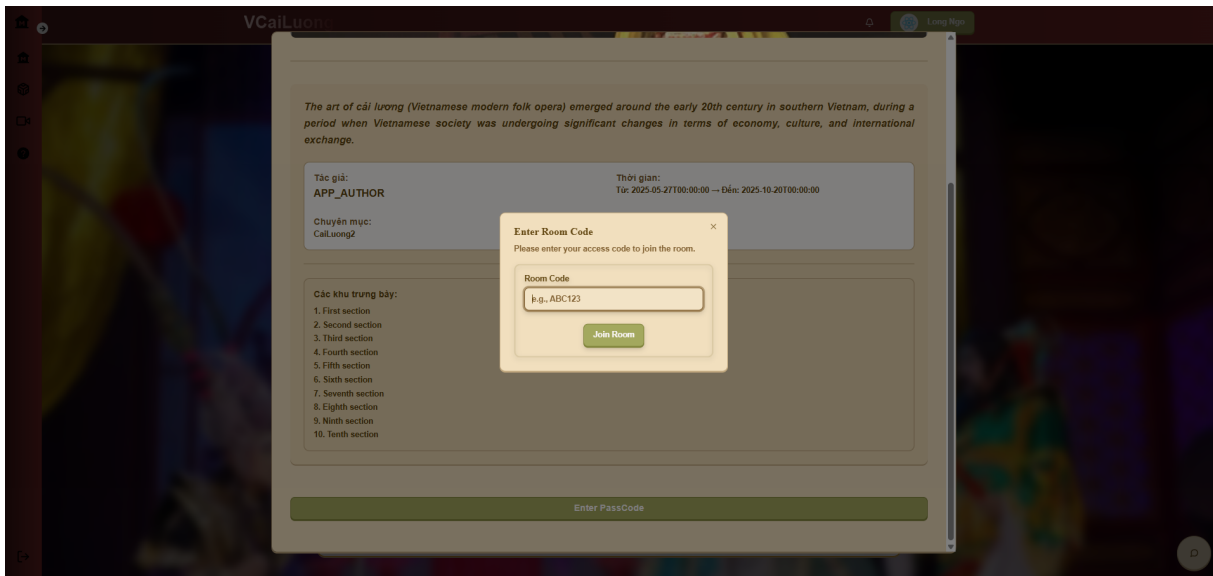


Figure 4.15: Entering Access Code UI in the VCaiLuong Visitor Application.

## 4.6.2 Museum Virtualization Implementation

The **Museum Virtualization** component is central to the VCaiLuong Visitor application, enabling the transformation of static 3D data into an interactive virtual museum environment. This process involves rendering the 3D structure of a museum and dynamically loading artefacts based on predefined configurations, allowing users to navigate and interact with exhibits in a way that mirrors real-world museum experiences.

To support this functionality, the system defines a structured database schema: each **Museum** entity is linked to a **Category**, which references a Template containing the 3D model and metadata. Templates consist of multiple **Sections**, each comprising several **Layouts** that serve as placeholders for artefacts.

The virtualization workflow is implemented as a sequential, four-step pipeline that progressively retrieves and processes data to render a fully interactive museum space. This approach not only enhances user engagement but also contributes to the preservation and digital promotion of Cai Luong cultural heritage.

### a) Fetching Museum Details

The initial phase involves retrieving the core museum data, including its associated category. The implementation steps are outlined as follows:

- The client-side application issues an API request to fetch museum details.
- The request is routed through the **API Gateway** to the *Visitor Service*.

- The *Visitor Service* constructs and forwards a proxy request to the *Core Service* to retrieve the required museum data.
- Upon receiving the data, the *Visitor Service* processes and returns the museum information as a response to the client.

#### b) Retrieval and Processing of Museum Template for 3D Visualization

Following the acquisition of core museum information, the next phase involves retrieving the associated museum template and preparing the corresponding 3D model file for client-side rendering. This step enables the dynamic visualization of the museum environment. The implementation proceeds as follows:

- The client-side application sends a request, using the previously obtained museum information, to retrieve the museum template.
- This request is routed through the **API Gateway** to the *Visitor Service*.
- The *Visitor Service* processes the request by querying the database to locate and retrieve the associated museum template record.
- Once the template record is retrieved, the service extracts the URL of the **.gltf** file from the template, which is stored in **Azure Blob Storage**. The **.gltf** file serves as the primary 3D model format, containing geometry, textures, and a binary **.bin** file.
- The service downloads the original **.gltf** file and parses it into a JSON object to allow for dynamic modification and cloud-access optimization.
- Within the JSON object, the **images** attribute, an array of texture file references, is examined. Each local URL is replaced with a **Shared Access Signature (SAS)** URL generated from Azure Blob Storage, providing secure, time-limited access.
- Similarly, the URI referencing the **.bin** file in the **.gltf** structure is replaced with an SAS URL for secure access to binary data.
- After completing all modifications, the updated JSON object is serialized back into a valid **.gltf** file format.
- The modified and cloud-accessible **.gltf** file is then streamed to the client application, enabling the rendering of the interactive 3D museum environment.

#### c) Retrieval and Preloading of Artifacts Based on Section Order

Once the core museum information has been obtained and the corresponding 3D template loaded, the system proceeds to retrieve artifacts associated with the museum. This is done by processing each **Section** in the defined order, ensuring the content is rendered consistently with the intended spatial and thematic layout. The implementation of this step follows a sequential and section-based approach:

- Based on the retrieved museum information, the system determines the ordered list of **Sections** defined within the template structure.
- For each section, the client-side application sends a request to fetch artifact data (referred to as *museum objects*) associated with that section.
- This request is routed through the **API Gateway** to the *Visitor Service*.
- The *Visitor Service* processes the request by querying the database to retrieve a list of artifacts and their associated layout configurations for the specified section. During this process, the service also generates **Shared Access Signature (SAS)** URLs for each artifact, enabling secure and time-limited access to media stored in **Azure Blob Storage**.
- The artifact data, along with the generated SAS URLs, is returned to the client in the form of an array of object values.
- Upon receiving the data, the client-side application preloads all related assets (e.g., images, videos, and 3D models) for the section using the provided SAS URLs from

Azure Blob Storage.

- This process is repeated for each section in the defined sequence. Once all sections and their artifacts have been preloaded, the application transitions into the **museum render state**, where the complete 3D virtual museum is visualized and made interactive for the user.

#### d) Rendering of Museum Structure and Textures

Once all artifact assets for the respective sections have been successfully preloaded, the system proceeds to render the virtual museum environment. This step involves reconstructing the museum's 3D structure and populating it with the corresponding artifacts.

- The client-side application utilizes the modified `.gltf` file, previously streamed from the server, to render the structural model of the museum.
- Following the structural rendering, the application maps preloaded artifact files (such as images, videos, and audio) to specific layout placeholders within the 3D environment. This mapping is performed by matching layout identifiers using the layout name as a key.
- For each layout, the associated artifact is dynamically assigned and embedded into the virtual space, ensuring correct spatial positioning and content presentation.
- Upon completing the rendering of all artifacts within the museum structure, the client application finalizes the 3D visualization process.
- Finally, the client issues a request to the server to log the user's room entry event, marking the user's participation and presence within the virtual museum environment. This event logging supports analytics and session tracking functionality within the system.

---

#### Algorithm 1 GetMuseumTemplate

---

**Input:** *museumInfo* – Object containing metadata of the selected museum

**Output:** Streamed GLTF file with cloud-based texture and binary references

```
1: function GETMUSEUMTEMPLATE(museumInfo)
2:   templateRecord  $\leftarrow$  Database.queryByMuseumId(museumInfo.id)
3:   gltfFileUrl  $\leftarrow$  templateRecord.templateFileUrl
4:   originalGLTF  $\leftarrow$  AzureBlob.download(gltfFileUrl)
5:   gltfObject  $\leftarrow$  PARSEJSON(originalGLTF)
6:   for all image in gltfObject.images do
7:     localUri  $\leftarrow$  image.uri
8:     sasUri  $\leftarrow$  AzureBlob.generateSAS(localUri)
9:     image.uri  $\leftarrow$  sasUri
10:  end for
11:  binUri  $\leftarrow$  gltfObject.buffers[0].uri
12:  sasBinUri  $\leftarrow$  AzureBlob.generateSAS(binUri)
13:  gltfObject.buffers[0].uri  $\leftarrow$  sasBinUri
14:  modifiedGLTF  $\leftarrow$  CONVERTToGLTF(gltfObject)
15:  return STREAMTOCLIENT(modifiedGLTF)
16: end function
```

---

---

**Algorithm 2** LoadMuseumArtifacts

---

**Input:** *museumInfo* – Object containing structured section information

**Output:** All artifact assets preloaded and ready for rendering

```
1: function LOADMUSEUMARTIFACTS(museumInfo)
2:   sectionList ← museumInfo.sections           ▷ Ordered list of museum sections
3:   for all section in sectionList do
4:     sectionId ← section.id
5:     artifactResponse ← APIClient.requestArtifacts(sectionId)
6:     artifactList ← artifactResponse.objects
7:     for all artifact in artifactList do
8:       PRELOADASSET(artifact.assetUrl)
9:     end for
10:  end for
11:  SETAPPLICATIONSTATE('READY_FOR_RENDER')
12: end function
```

---



Figure 4.16: Virtual Museum in the VCaiLuong Visitor Application.

### 4.6.3 Implementation of Interactive Features

In addition to virtualizing museum environments, the **VCaiLuong Visitor system** enables rich user interaction within the virtual museum room. This includes both personal engagement with artifacts and navigation, as well as social interaction with other users, fostering immersive exploration and collaborative learning around Cai Luong heritage.

#### a) Interaction with the Museum Environment and Artifacts

The primary level of interactivity in the **VCaiLuong Visitor system** centers on user engagement with the virtual museum space and its embedded artifacts. Designed to resemble a game-like experience, users navigate the 3D environment using standard key-

board controls (**W**, **A**, **S**, **D**) for directional movement and mouse input for interactions.

Following the completion of the museum’s 3D rendering, the application presents an introductory interface that outlines the current museum’s context and provides user guidance for in-room navigation and interaction rules. Once inside, users can explore the space freely.

When an artifact is clicked, a contextual pop-up displays detailed, preloaded information about the object. By relying on data fetched during the initial loading phase, the system avoids additional network requests during runtime, thereby ensuring low-latency performance and uninterrupted 3D rendering.

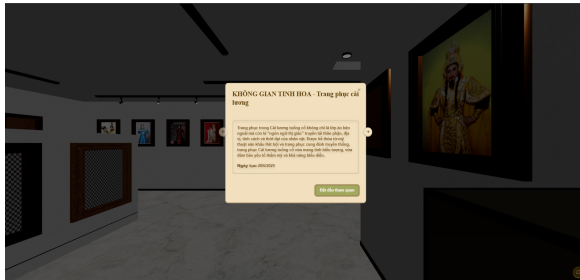


Figure 4.17: Virtual Museum Introduction in the VCaiLuong Visitor Application.



Figure 4.18: Artifact Description in the VCaiLuong Visitor Application.

### b) Text-to-Speech Integration for Enhanced Accessibility

To enhance user experience and accessibility, artifact information popups in the VCaiLuong Visitor system include a Text-to-Speech (TTS) feature. Powered by Azure Speech Service, this functionality allows users to listen to artifact descriptions in various voices and languages. When a user clicks the "Play" button, the client sends a request with authentication details and a language code to Azure. After validation, Azure returns an audio file that plays automatically, providing an inclusive and seamless auditory experience within the virtual museum.

### c) Interaction between other users through the chat box

Upon successful entry into the 3D museum environment, the VCaiLuong system enables real-time communication among users through a chat box feature implemented using WebSocket technology. This functionality allows users within the same virtual museum room to exchange messages instantly, thereby supporting collaborative exploration and cultural dialogue.

#### WebSocket Connection Establishment And Disconnect:

- The client initiates a `joinRoom` event via WebSocket to join the chat session for the current museum room.
- The server receives the event, verifies the room context, and authorizes the connection.
- Upon successful validation, the server registers the user in a room-specific channel for real-time communication.
- A persistent WebSocket connection is maintained, allowing continuous, bidirectional message exchange without repeated HTTP calls.
- When the user exits the 3D museum, the client emits a `leaveRoom` event.
- The server removes the user from the room channel, stopping further message delivery.

#### Real-Time Messaging Workflow:

Once the **WebSocket** connection is established, real-time message exchange is executed through an event-based communication model:

- When a user sends a message, the client emits a **chatToServer** event containing the message content and metadata such as the sender's ID and room ID.
- The server processes the message and broadcasts it to all other users in the same room by emitting a **chatToClient** event.
- Clients subscribed to the **chatToClient** event display the incoming message in the user interface, enabling real-time communication.

This event-driven system enables synchronous interaction while avoiding the performance penalties associated with traditional polling mechanisms. It is optimized to support an interactive and engaging user experience inside the **VCaiLuong virtual museum**.

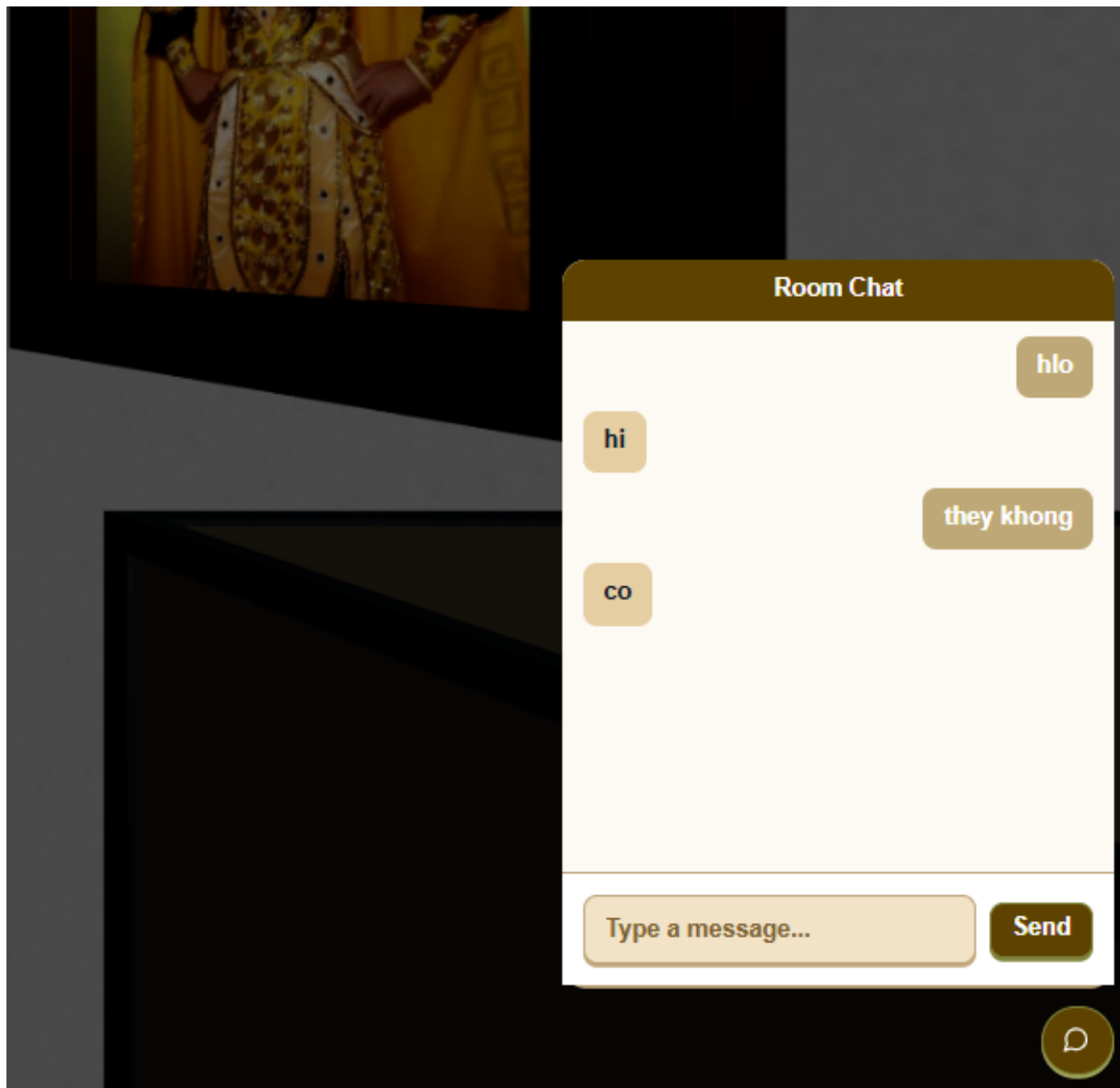


Figure 4.19: Real Time Chat Box In Virtual Museum.

## 4.7 Watch Party Feature Implementation

In addition to museum virtualization, the **VCaiLuong system** offers an interactive Watch Party feature that allows users to view recorded Cai Luong performances, ranging from documentary films to live performance recordings, within a shared virtual environment. This implementation seamlessly combines client-side operations, server-side services, and **WebSocket-based** communication to provide a synchronized and socially engaging viewing experience.

### 4.7.1 Retrieving and Streaming Video Content

The Watch Party experience begins with the client initiating a request to retrieve information about an available Watch Party session. This includes metadata such as the video title, description, and streaming configuration.

- The client application sends an API request to fetch Watch Party details.
- The request is routed through the API Gateway and forwarded to the Visitor service.
- The Visitor service queries the database to retrieve the relevant Watch Party record, which includes the reference to the video file.
- Using this reference, the server generates a Shared Access Signature (SAS) URL for the video file hosted on Azure Blob Storage.
- The SAS URL, along with the other Watch Party metadata, is returned to the client as part of the response.
- Upon receiving the SAS URL, the client preloads the video content from Azure Blob Storage and automatically begins playback once the preload is successfully completed.

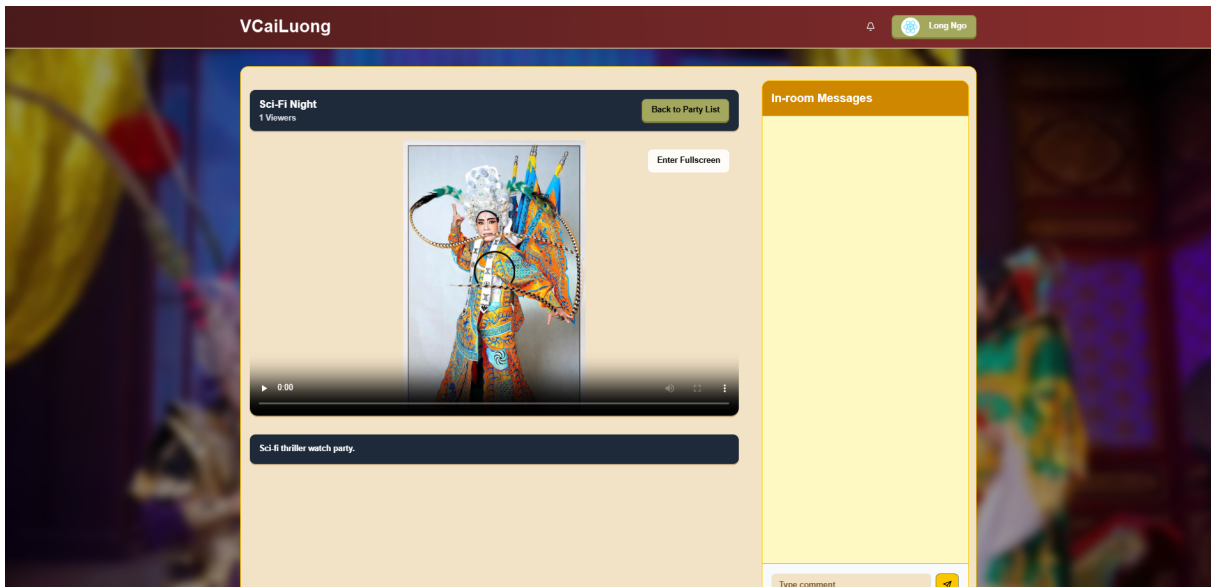


Figure 4.20: Watch Party in VCaiLuong Visitor Application.

### 4.7.2 Enabling Real-Time Interaction via WebSocket

To foster user interaction during the Watch Party session, the system includes a real-time chat feature implemented through WebSocket technology. This allows viewers to



communicate with each other while watching the same performance.

- When a user joins a Watch Party session, the client emits a `joinRoom` event via WebSocket to establish a connection with the server.
- The server accepts the connection, registers the user to the room, and logs the entry for session tracking and engagement analytics.
- When a user sends a message, the client emits a `chatToServer` event, including the message content and session metadata.
- The server processes the message and broadcasts it to all users in the same room using the `chatToClient` event.
- Clients subscribed to the `chatToClient` event display the incoming message in their chat interface.
- When the user leaves the session, the client emits a `leaveRoom` event.
- The server removes the user from the room's communication channel and logs the departure for accurate activity tracking.

This architecture ensures low-latency communication and synchronous chat interaction, significantly enhancing the collaborative viewing experience. It mirrors the social aspects of physical gatherings by allowing real-time commentary and dialogue during cultural content playback.

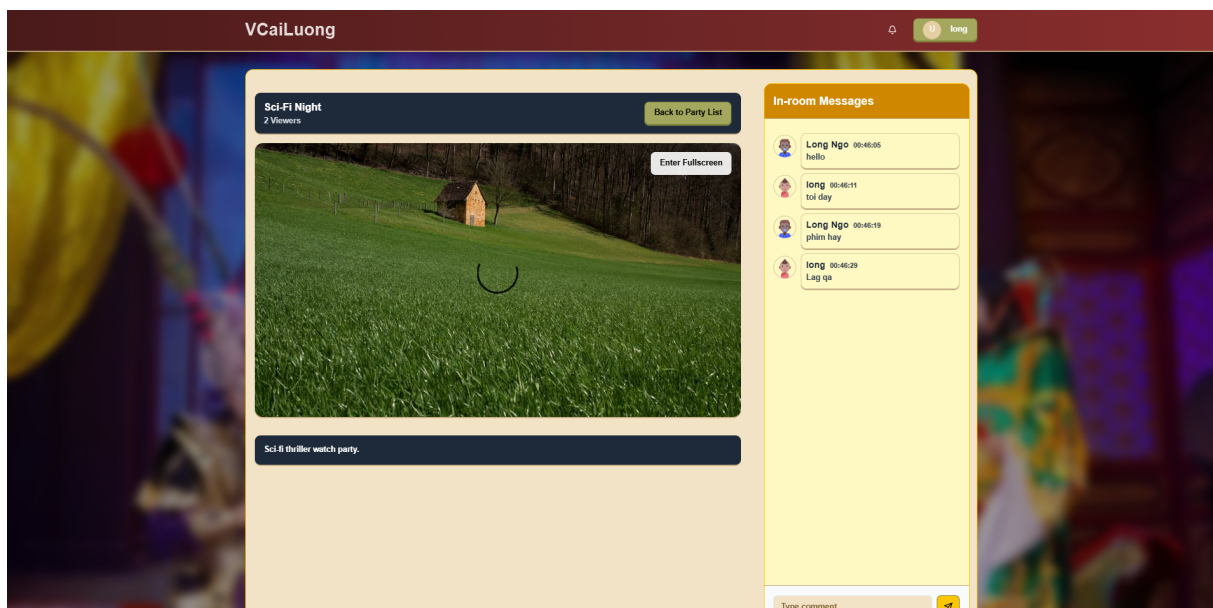


Figure 4.21: Watch Party Chat Box UI.



# Chapter 5

## DISCUSSION

### 5.1 Discussion

The VCaiLuong System serves as a unified platform dedicated to the preservation of Cai Luong and other related traditional art forms. By offering a modern and immersive approach to engaging with this intangible cultural heritage, the system bridges the gap between historical traditions and contemporary audiences, particularly the younger generation.

In addition to cultural storytelling, VCaiLuong provides a digital archive for physical artifacts, transforming them into structured digital records. These records are then exhibited within a virtual museum environment, making them accessible to a broader community. This approach not only safeguards valuable artifacts against deterioration over time but also presents them as part of a curated narrative that reflects the historical and artistic richness of Vietnamese traditional performing arts.

### 5.2 Strengths

**Virtualization of the Museum:** The VCaiLuong system offers a virtual 3D museum experience via a web-based application. This immersive environment enhances user engagement and effectively promotes Cai Luong culture to younger audiences who are more inclined toward digital interaction.

**User Interactions:** The system integrates multiple interactive features such as chat boxes and threaded comment sections. These tools foster real-time communication and collaboration among users, creating a more connected and engaging cultural learning environment.

**Community-Oriented Cultural Events:** Through the Watch Party feature, VCaiLuong enables visitors to participate in shared viewing sessions of live Cai Luong performances, documentary films, and workshops. This functionality not only supports cultural education but also fosters a sense of community through collective discussion and reflection.

**Automated Notifications and User-Centric Workflows:** To improve accessibility and user convenience, the system includes automated features such as notifications and booking support. These tools allow users to track upcoming museum events, receive timely access codes, and prepare in advance for their visits, thereby enriching their cultural experience.

**Cloud-Based Artifact Storage:** VCaiLuong operates as a cloud-integrated platform that utilizes modern cloud computing technologies. By leveraging services such as Azure Blob Storage, the system ensures secure, scalable, and durable storage for digitized cultural artifacts. This digital transformation supports long-term preservation and facilitates future accessibility.

**Scalability and Integration Capabilities:** The system is designed with modularity in mind, utilizing a microservices architecture to support extensibility. Its integration with cloud services allows for seamless deployment of new features without disrupting

existing functionalities. In addition, cloud-based solutions enable real-time monitoring and application scaling to handle increased traffic. Performance is further optimized through the use of caching mechanisms and database tuning strategies.

## 5.3 Limitations

**User Support and Notification Configuration:** While the VCaiLuong system provides foundational features for user interaction and engagement, there is a need to further enhance user support functionalities. In particular, the current notification system is limited in scope. Future improvements should include support for multiple notification types and personalized settings. Additionally, scheduled jobs within the system should be better managed to avoid overlapping tasks that could exhaust system resources or degrade performance.

**Performance Optimization:** Certain features of the system, most notably the loading and rendering of virtual museums, require further optimization to improve overall user experience. Delays in rendering or latency in asset retrieval can negatively affect user immersion, particularly in resource-constrained environments or under high traffic loads. Future work should focus on refining backend query performance, front-end rendering efficiency, and asset streaming strategies.

**User Interface Enhancements:** Although the current user interface is functional, it can benefit from visual and usability improvements to appeal more effectively to younger audiences. At the same time, these enhancements should retain the cultural identity and traditional aesthetic of the Cai Luong heritage. A more intuitive, responsive, and visually engaging UI would help attract a broader user base while respecting the historical and artistic value of the content.

## 5.4 Comparison With Other Systems

The VCaiLuong system is inspired by the goals and functionalities of both the VNFAM and DHC systems. Drawing from their approaches to cultural preservation and digital engagement, VCaiLuong builds upon their foundations with a focus on virtual interactivity and user accessibility.

A comparison among the three platforms highlights both the achievements and limitations of VCaiLuong. While it excels in areas like 3D visualization, user interaction, and cloud-based artifact storage, this comparison also reveals opportunities for further enhancement, particularly in system performance, administrative tools, and user experience design.

Table 5.1: Comparison of DHC, VNFAM, and VCaiLuong Systems

| Aspect                       | DHC System                                                | VNFAM System                                         | VCaiLuong System                                                                            |
|------------------------------|-----------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Visualization</b>         | Basic web-based interface with traditional layouts        | Immersive 3D virtual tours with panoramic rendering  | Interactive 3D virtual museum with artifact placement, animated navigation                  |
| <b>User Functionality</b>    | Community content contribution and public discourse tools | Limited user interaction, focused on curated viewing | Rich interaction including 3D navigation, artifact pop-ups, Watch Party, live chat          |
| <b>Cultural Preservation</b> | Emphasizes intangible heritage (rituals, oral traditions) | Digitizes tangible artifacts (paintings, objects)    | Combines tangible and intangible heritage through narration, media, and interactive content |
| <b>Scalability</b>           | Less scalable, limited to web archives                    | Moderate scalability with curated backend            | Cloud-native design with microservices and caching for high scalability                     |
| <b>Artifact Storing</b>      | Basic file storage with limited media support             | Centralized storage for curated assets               | Cloud-based multi-media storage using Azure Blob and dynamic SAS URL access                 |

## 5.5 Future Work

Future development of the **VCaiLuong system** will focus on enhancing both functionality and performance. The UI of the **Visitor and Admin applications** will be improved to ensure a more modern, accessible, and engaging experience, especially for younger users. Integration of AI features, such as intelligent search, content recommendations, and natural language interaction, will further personalize the cultural exploration process. Additionally, backend optimization efforts will address system performance, scalability, and resource efficiency to support increased traffic and reduce latency.

The **Admin application** will also be expanded with features such as content approval workflows, analytics dashboards, and system monitoring tools. A key enhancement includes the ability to create and configure virtual museums directly from the Admin interface, allowing for the easy setup of exhibition layouts, artifact uploads, and interactive elements. These improvements will empower administrators and curators to manage content efficiently while ensuring seamless integration with the **Visitor application**.

# Chapter 6

## CONCLUSION

The development and deployment of the **VCaiLuong system** mark a significant step forward in the digital preservation and promotion of Cai Luong, an important form of intangible cultural heritage in Vietnam. By leveraging modern web technologies, the system offers a multifaceted solution that combines cultural education, digital archiving, and interactive community engagement.

**VCaiLuong** creates a dynamic cultural space by virtualizing museums, allowing users to explore artifacts and stories through a 3D interactive platform. This approach not only ensures the long-term preservation of artifacts in digital format but also supports cultural transmission through immersive experiences. Features such as real-time chat, Watch Parties, and scheduled exhibitions provide users with new ways to engage with traditional content, making Cai Luong more accessible, especially to younger generations.

In addition, the platform offers tools that benefit institutions such as theaters and cultural research centers. These organizations can use **VCaiLuong** to display artifacts, host virtual events, and connect with a broader audience. This strengthens their ability to preserve and promote the art form while also supporting its continued development in the digital age.

Overall, **VCaiLuong** demonstrates how digital transformation can enrich cultural heritage preservation by bridging the gap between tradition and innovation.

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# Appendix A

## Source Code Repository

The full source code for the **VCaiLuong** system—including the Visitor and Admin applications, as well as backend services—is publicly available on GitHub. The project is divided into several repositories as follows:

- **Visitor Application (Frontend):** <https://github.com/NVCLong/Digital-Museum>
- **Visitor Service :** [https://github.com/NVCLong/Digital\\_Museum\\_BE](https://github.com/NVCLong/Digital_Museum_BE)
- **Common Backend Service:** [https://github.com/NVCLong/Common\\_BE](https://github.com/NVCLong/Common_BE)
- **Core Backend Service:** [https://github.com/NVCLong/Core\\_BE](https://github.com/NVCLong/Core_BE)

These repositories contain the full source code, documentation, and deployment configurations necessary to run and develop the VCaiLuong platform.